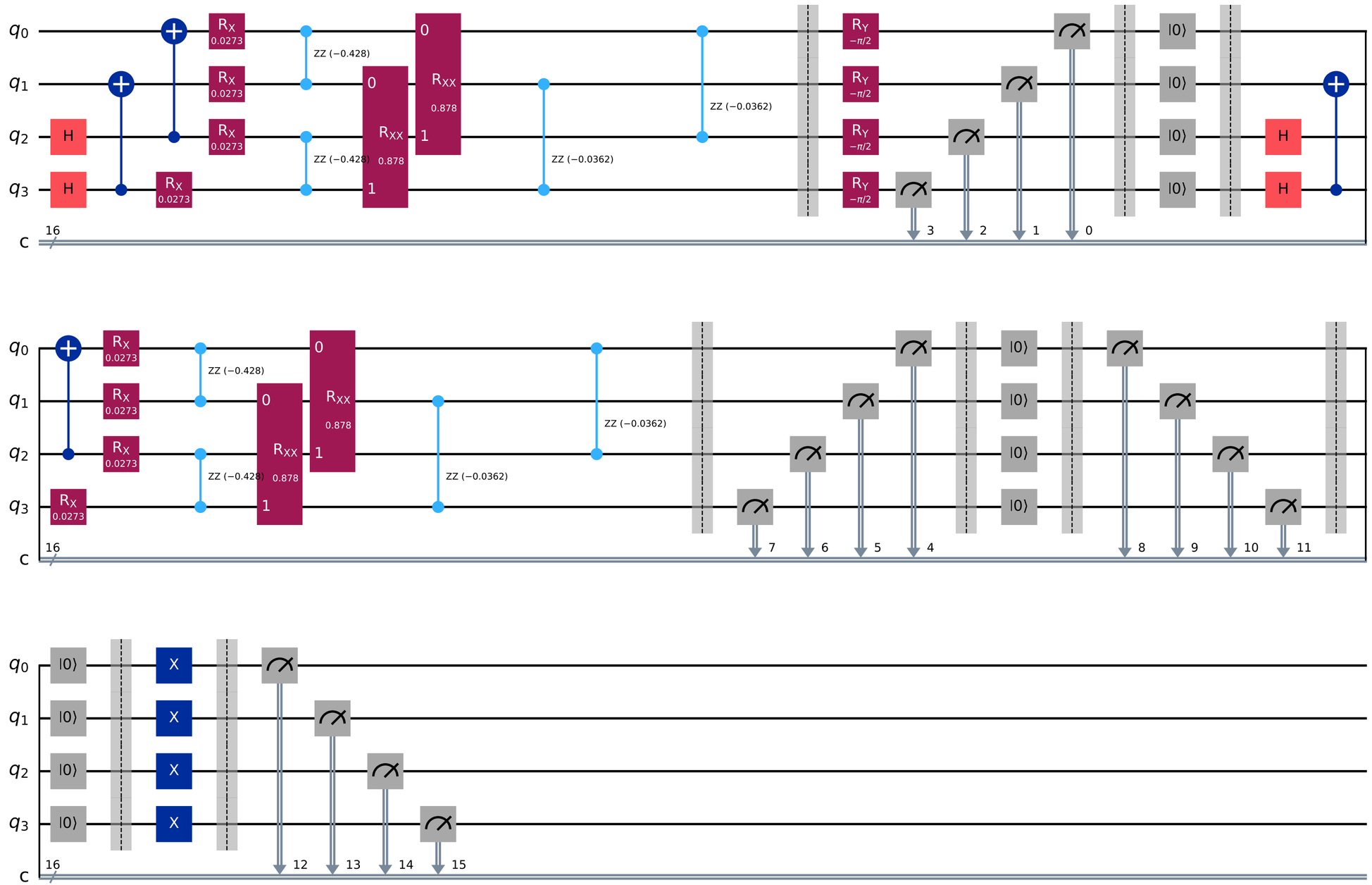
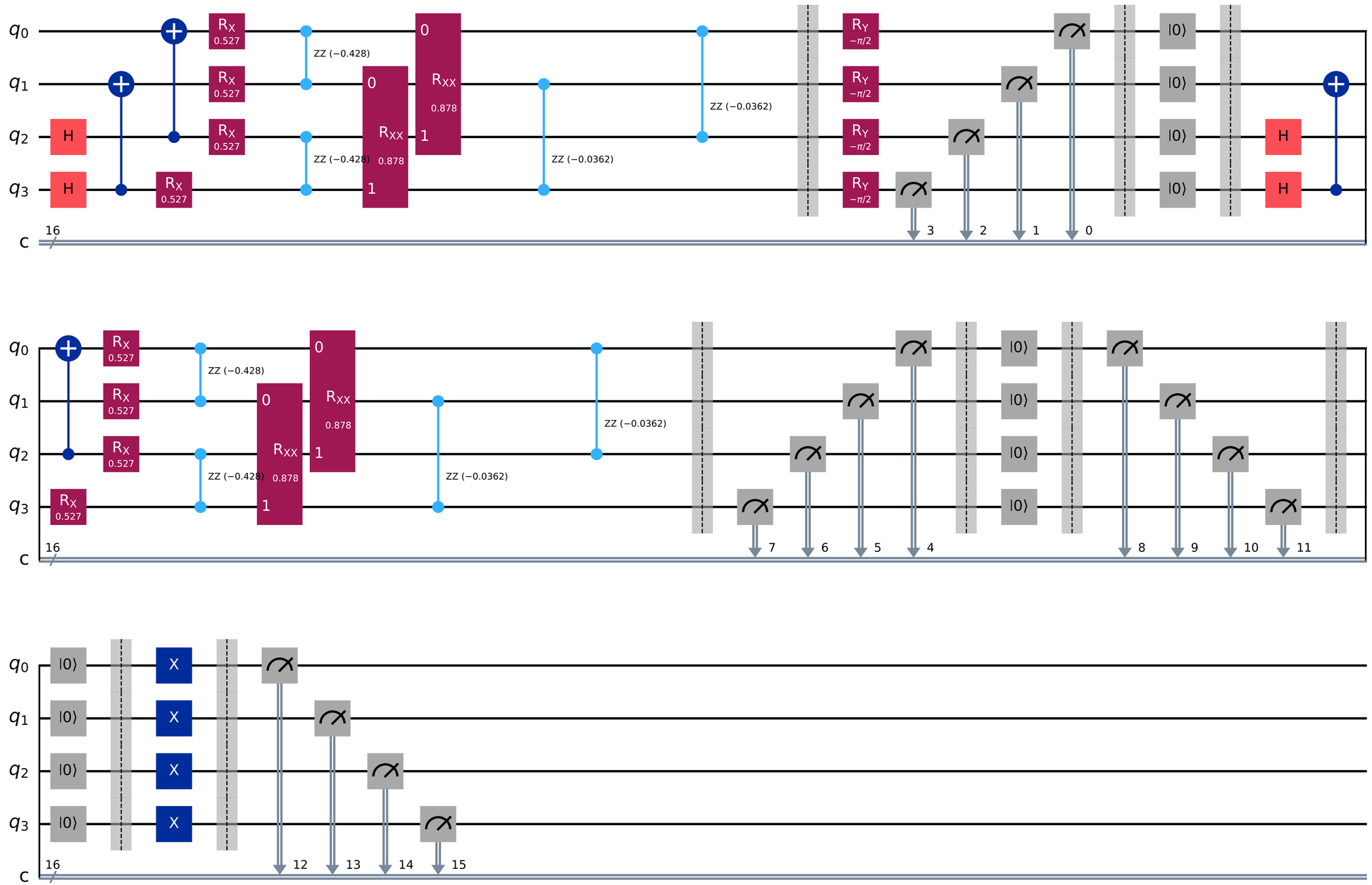
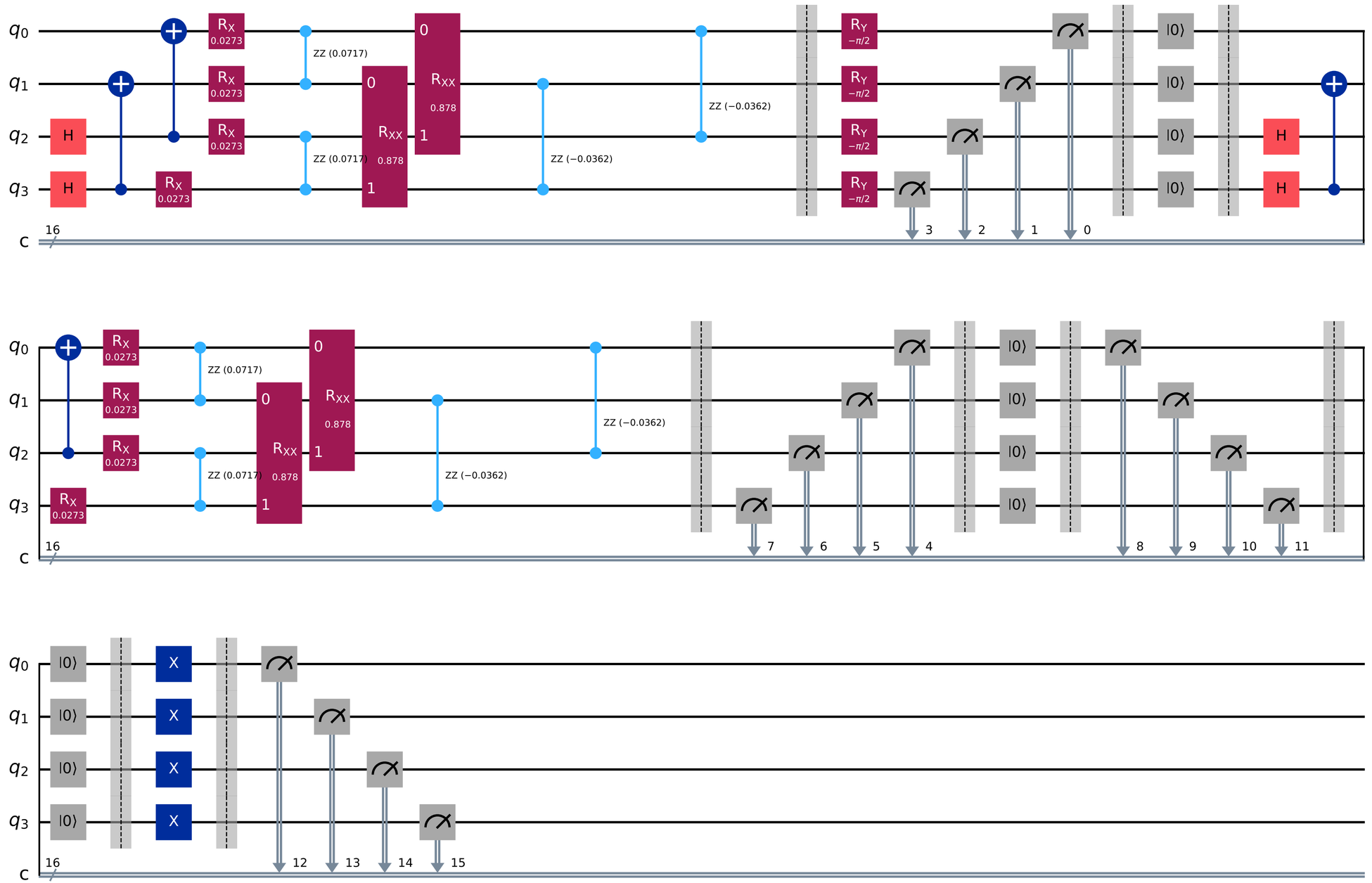
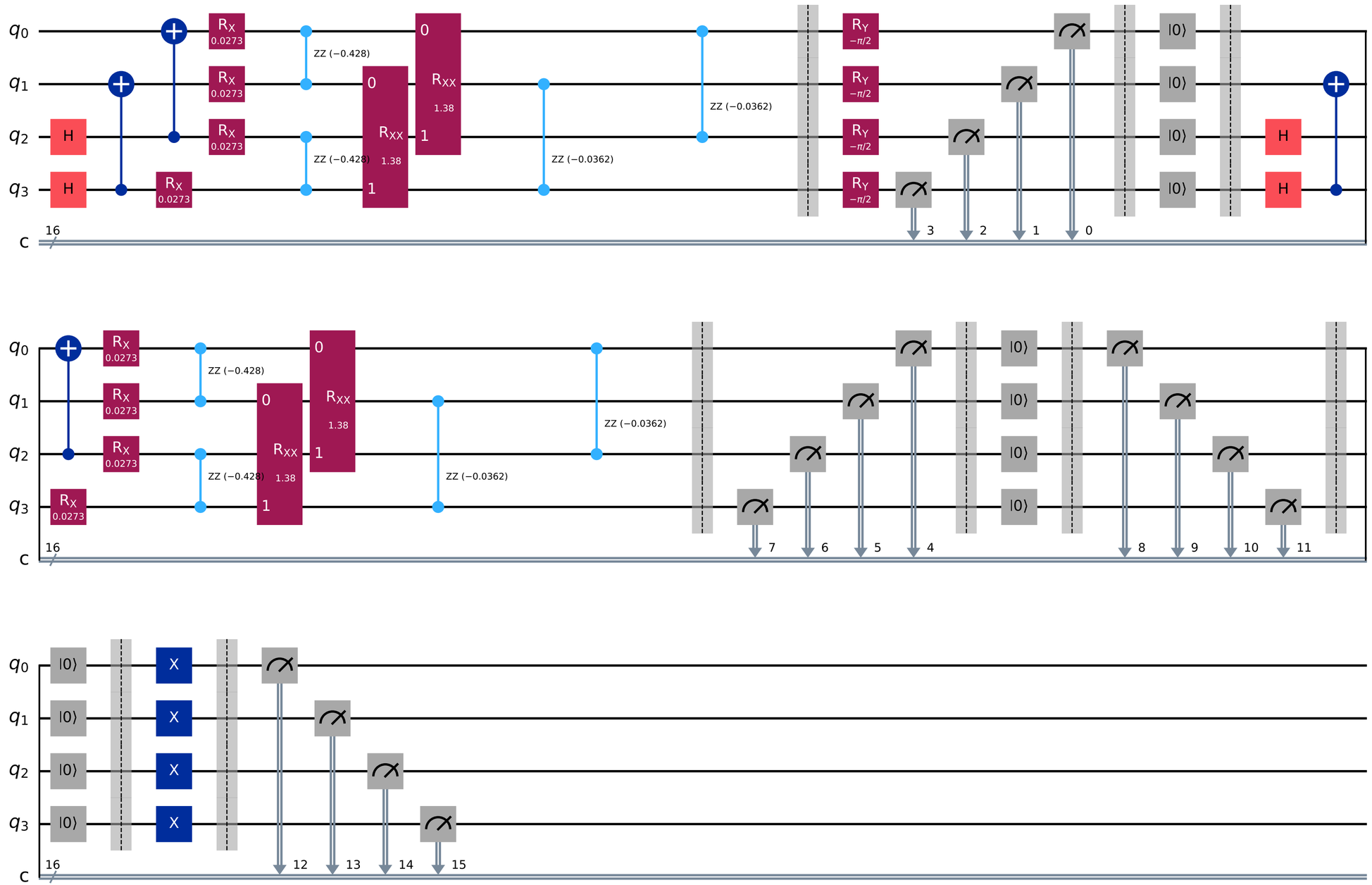
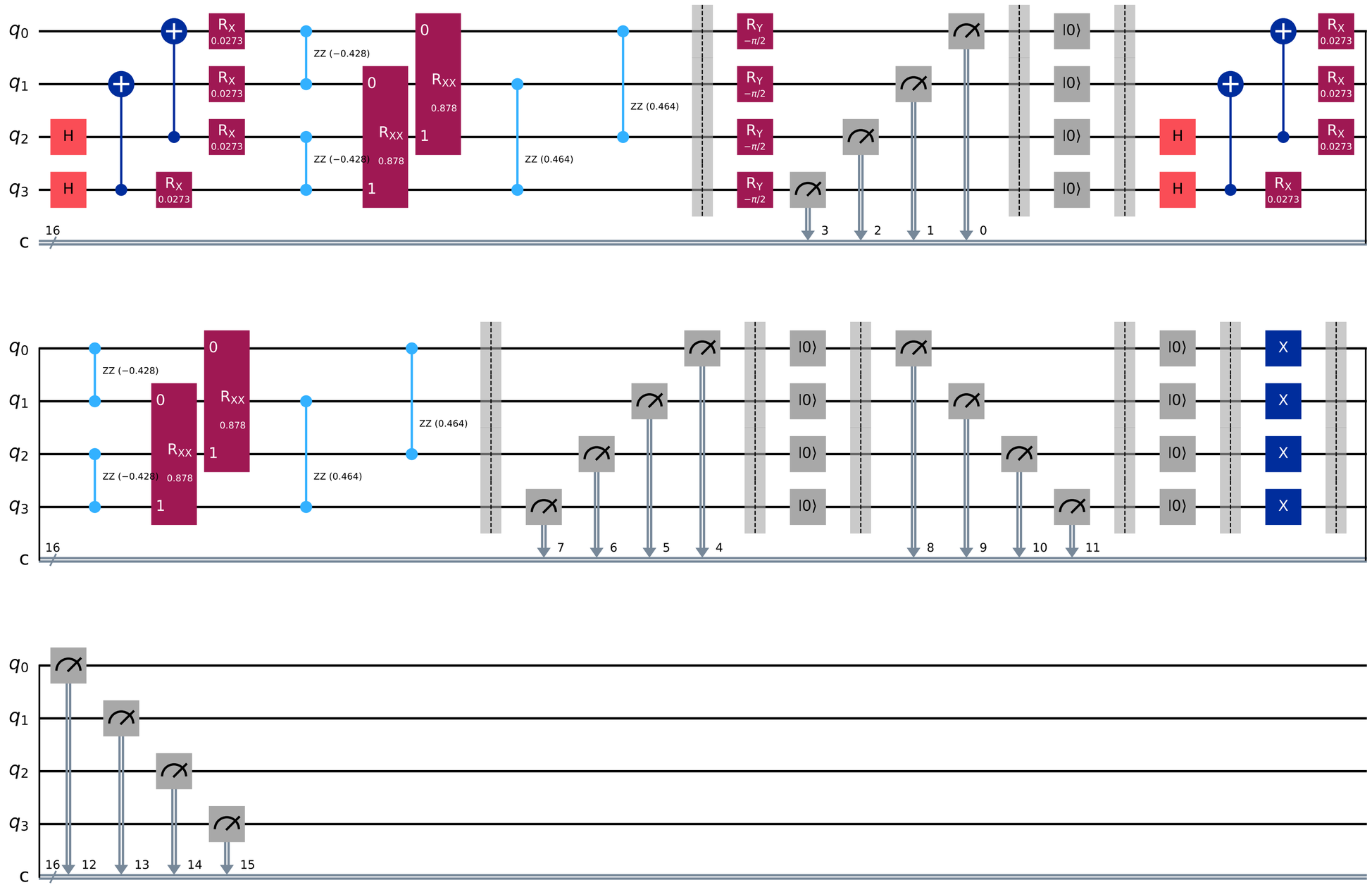
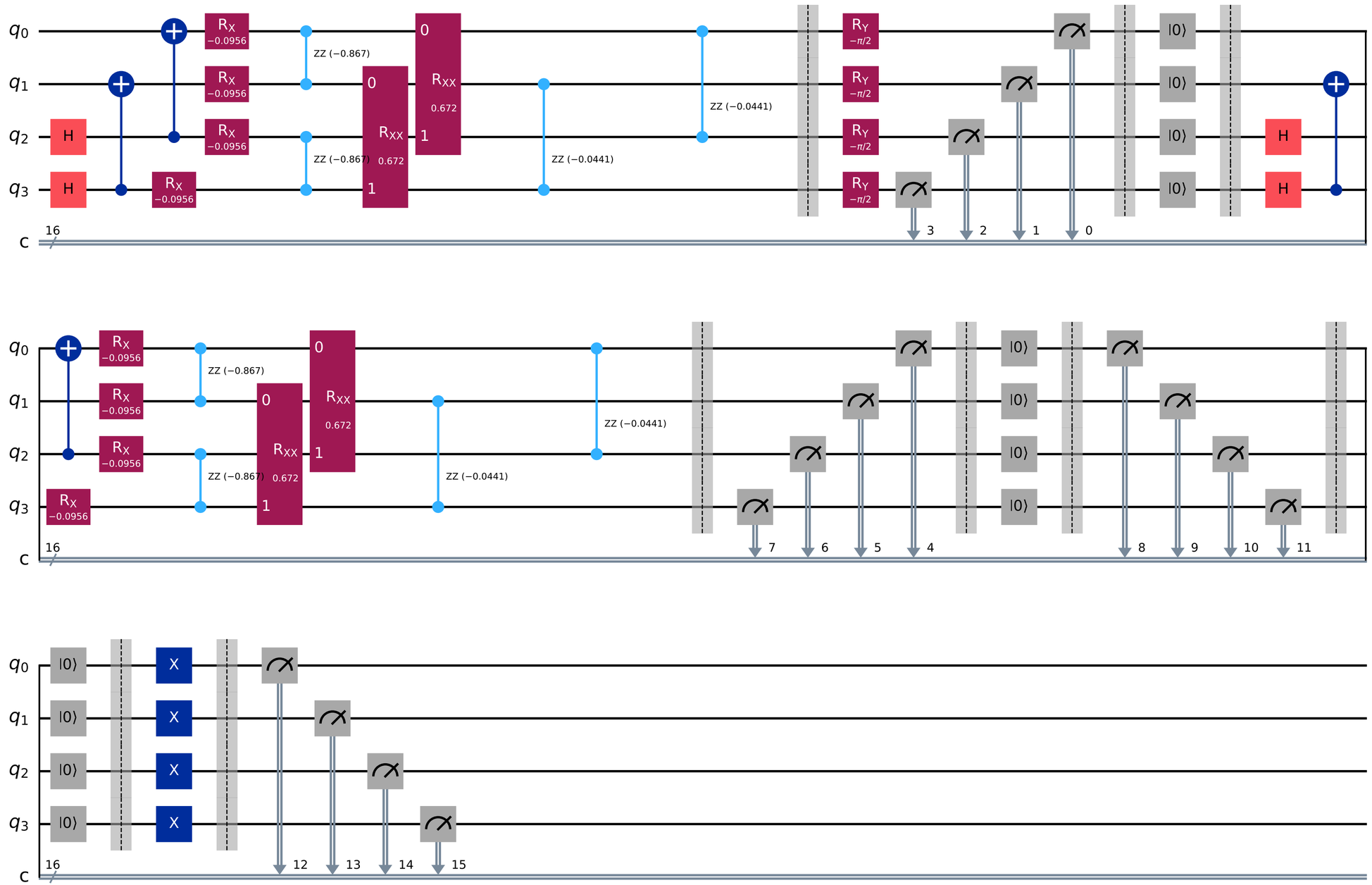

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0273
R_X 0.0273
R_X 0.0273
R_X 0.0273
ZZ (−0.428)
ZZ (−0.428)
0
R_{XX} 0.878
1
0
R_{XX} 0.878
1
ZZ (−0.0362)
ZZ (−0.0362)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X 0.0273
R_X 0.0273
R_X 0.0273
R_X 0.0273
ZZ (−0.428)
ZZ (−0.428)
0
R_{XX} 0.878
1
0
R_{XX} 0.878
1
ZZ (−0.0362)
ZZ (−0.0362)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

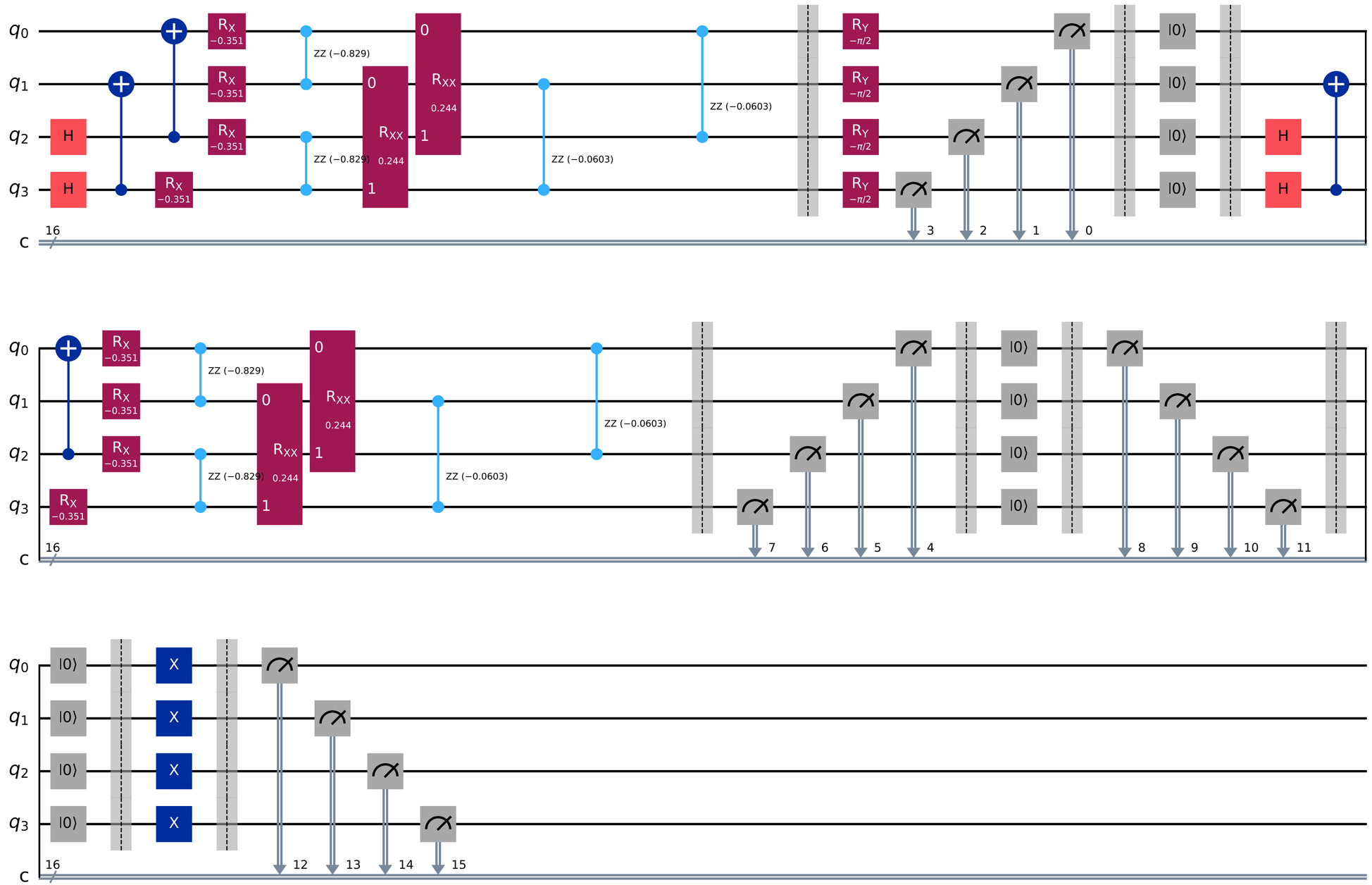

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.527
R_X 0.527
R_X 0.527
R_X 0.527
ZZ (−0.428)
ZZ (−0.428)
0
R_{XX} 0.878
1
0
R_{XX} 0.878
1
ZZ (−0.0362)
ZZ (−0.0362)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X 0.527
R_X 0.527
R_X 0.527
R_X 0.527
ZZ (−0.428)
ZZ (−0.428)
0
R_{XX} 0.878
1
0
R_{XX} 0.878
1
ZZ (−0.0362)
ZZ (−0.0362)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

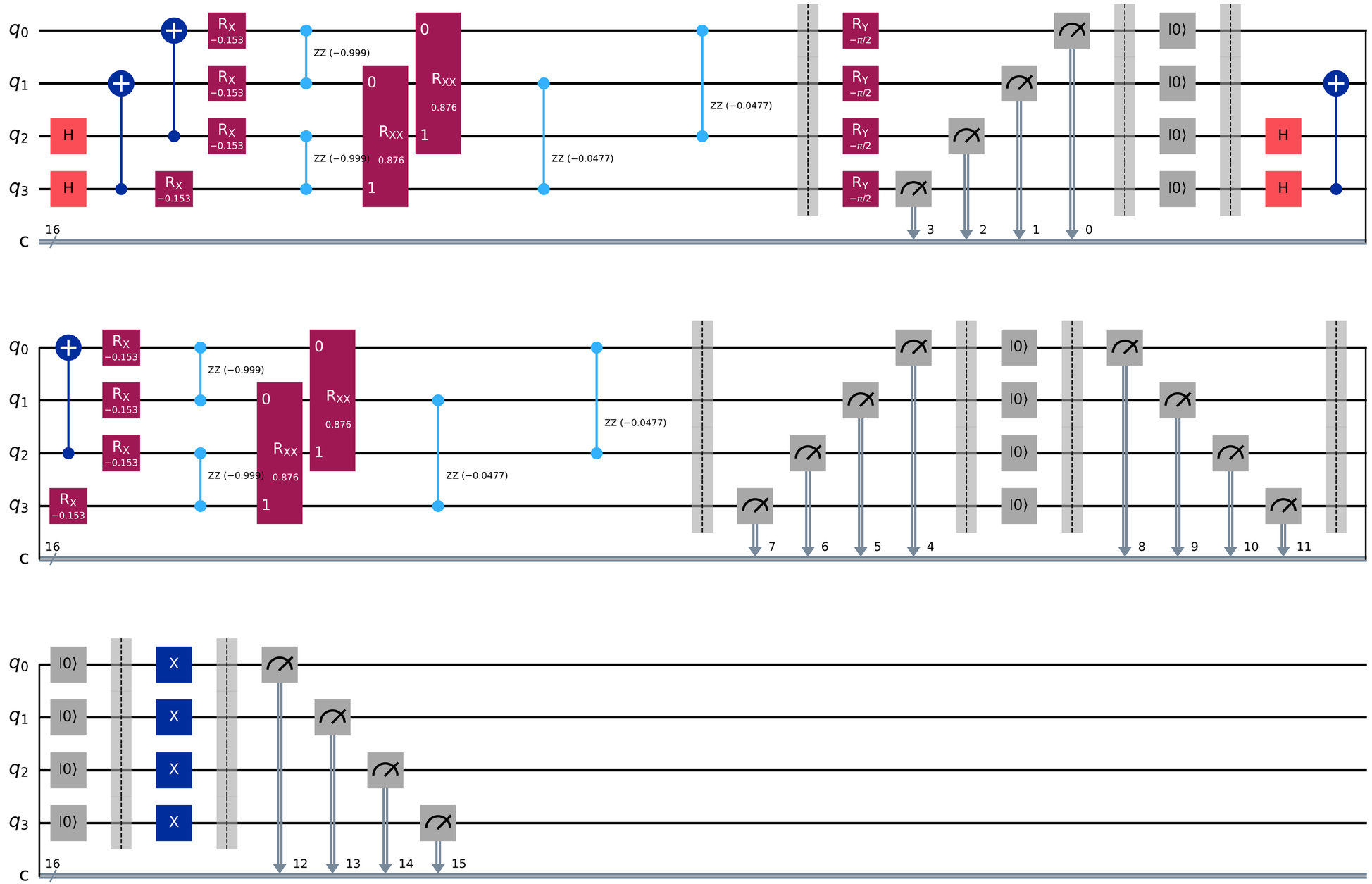

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0273
R_X 0.0273
R_X 0.0273
R_X 0.0273
ZZ (0.0717)
ZZ (0.0717)
0
R_{XX} 0.878
1
0
R_{XX} 0.878
1
ZZ (−0.0362)
ZZ (−0.0362)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X 0.0273
R_X 0.0273
R_X 0.0273
R_X 0.0273
ZZ (0.0717)
ZZ (0.0717)
0
R_{XX} 0.878
1
0
R_{XX} 0.878
1
ZZ (−0.0362)
ZZ (−0.0362)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

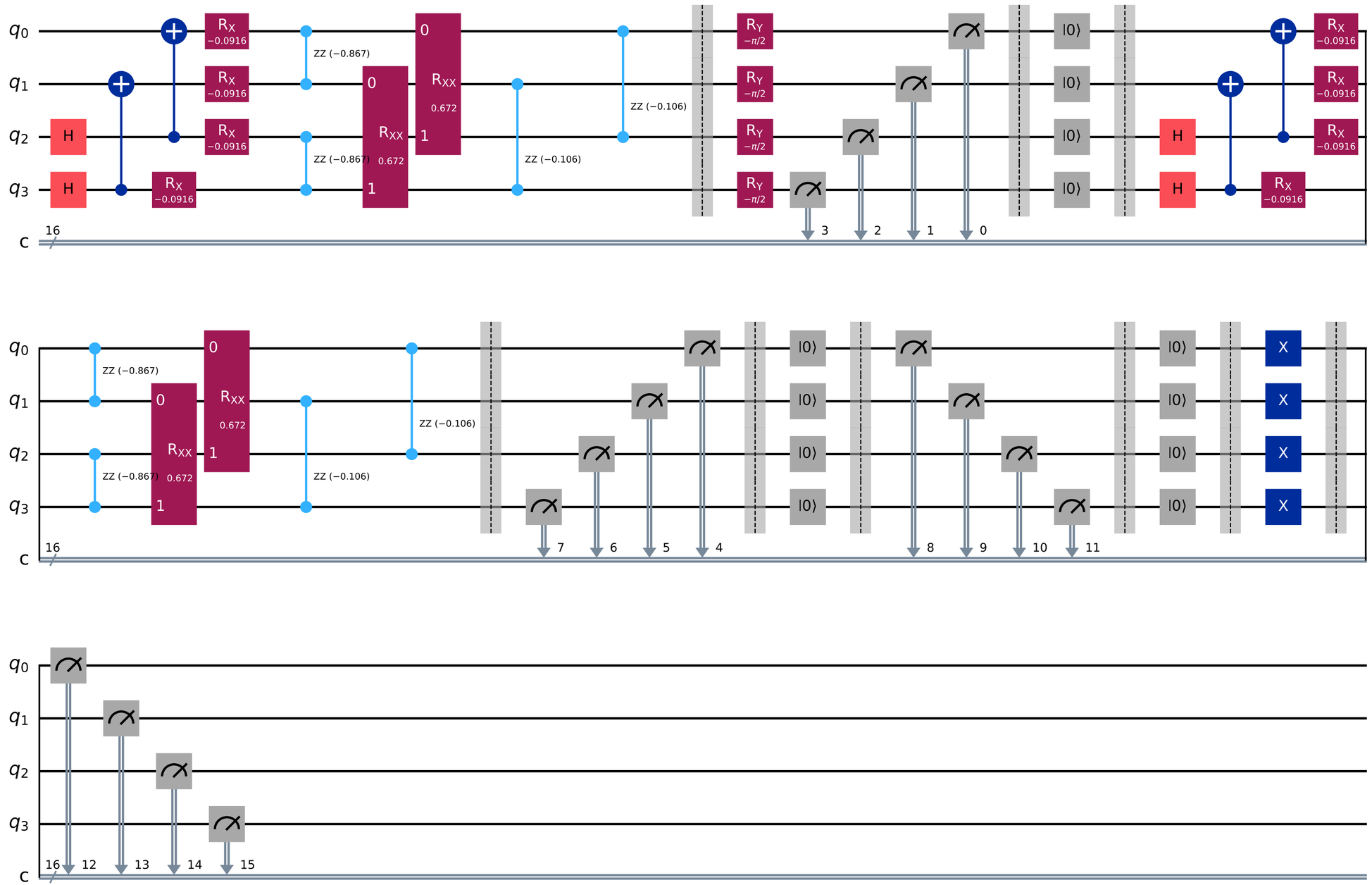

q0
q1
q2
q3
c
16
H
H
RX 0.0273
RX 0.0273
RX 0.0273
RX 0.0273
ZZ (−0.428)
ZZ (−0.428)
0
RXX 1.38
1
0
RXX 1.38
1
ZZ (−0.0362)
ZZ (−0.0362)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
RX 0.0273
RX 0.0273
RX 0.0273
RX 0.0273
ZZ (−0.428)
ZZ (−0.428)
0
RXX 1.38
1
0
RXX 1.38
1
ZZ (−0.0362)
ZZ (−0.0362)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

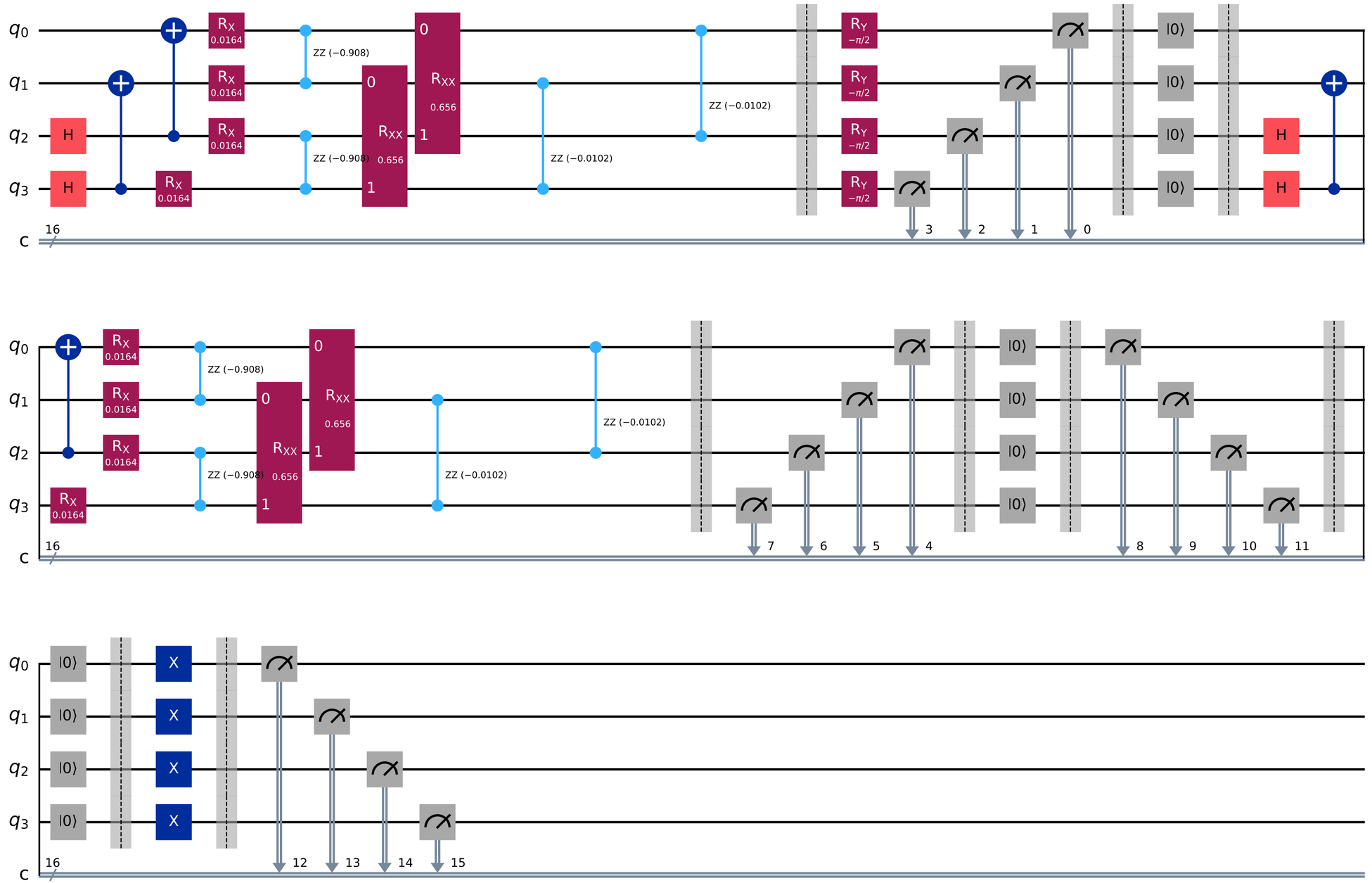

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0273
R_X 0.0273
R_X 0.0273
R_X 0.0273
ZZ (−0.428)
ZZ (−0.428)
0
R_{XX} 0.878
1
0
R_{XX} 0.878
1
ZZ (0.464)
ZZ (0.464)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.0273
R_X 0.0273
R_X 0.0273
R_X 0.0273
ZZ (−0.428)
ZZ (−0.428)
0
R_{XX} 0.878
1
0
R_{XX} 0.878
1
ZZ (0.464)
ZZ (0.464)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
q_0
q_1
q_2
q_3
c
16
12
13
14
15

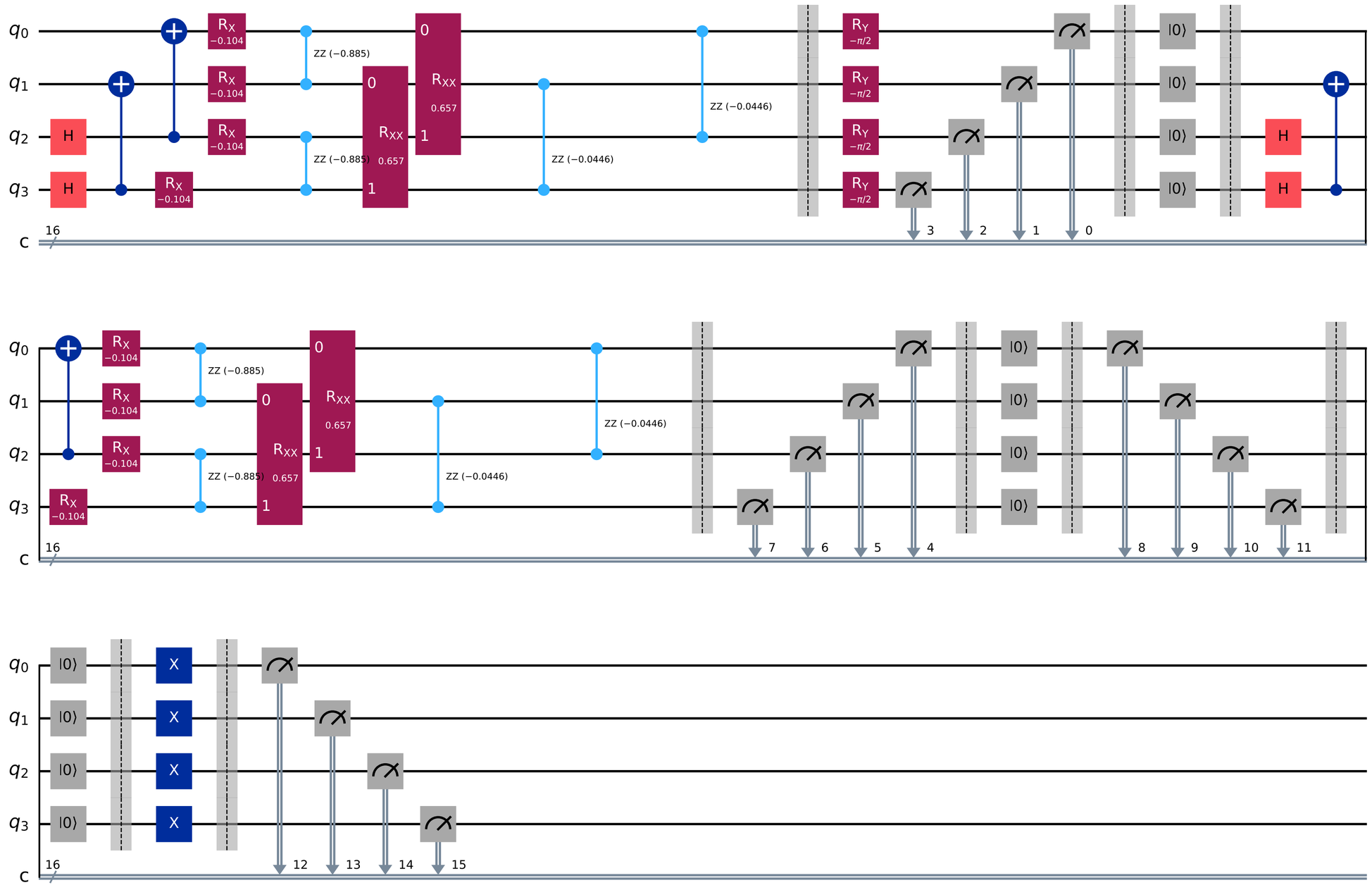

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0956
R_X −0.0956
R_X −0.0956
R_X −0.0956
ZZ (−0.867)
ZZ (−0.867)
0
R_{XX} 0.672
1
0
R_{XX} 0.672
1
ZZ (−0.0441)
ZZ (−0.0441)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.0956
R_X −0.0956
R_X −0.0956
R_X −0.0956
ZZ (−0.867)
ZZ (−0.867)
0
R_{XX} 0.672
1
0
R_{XX} 0.672
1
ZZ (−0.0441)
ZZ (−0.0441)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

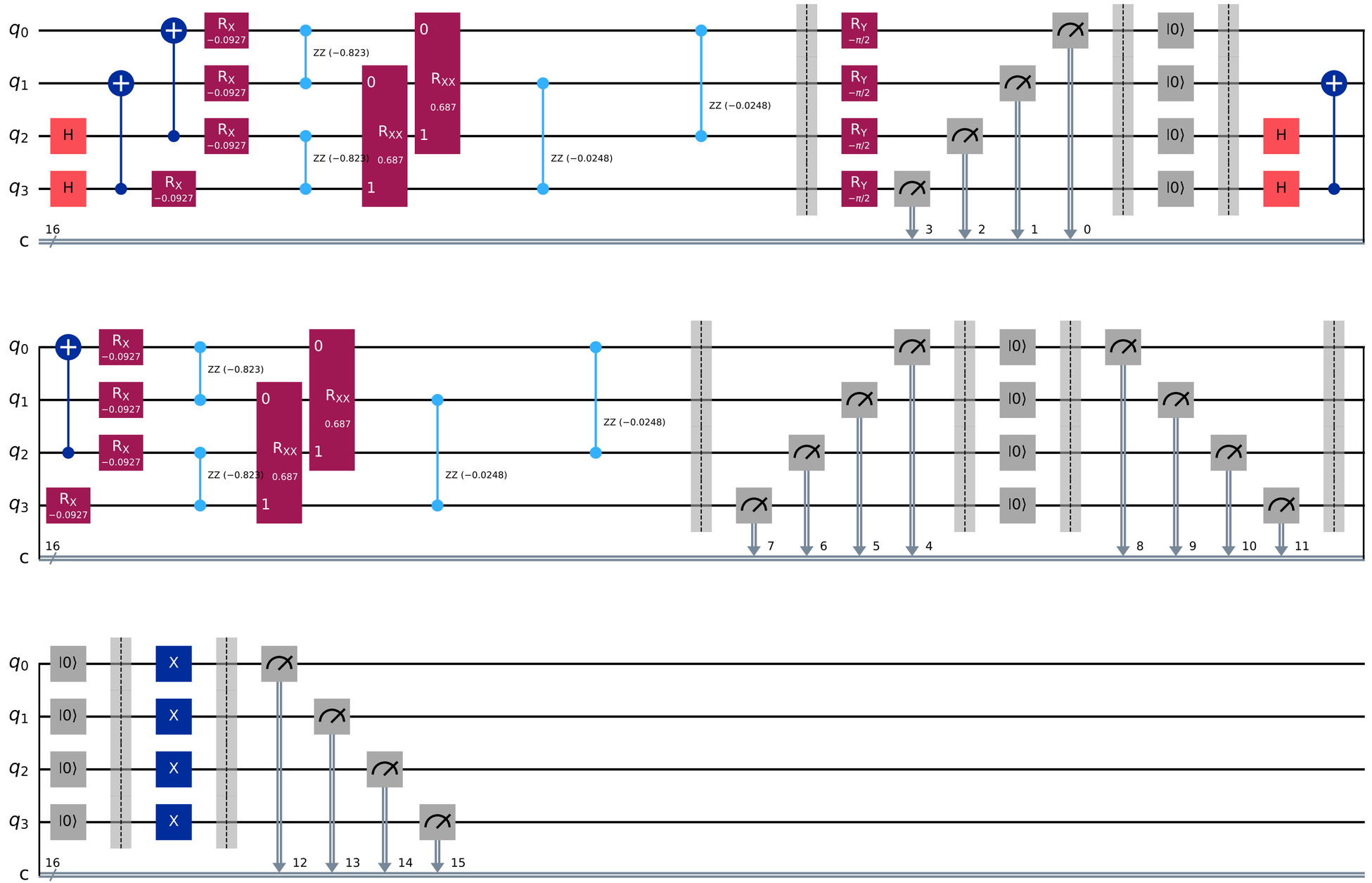

q0
q1
q2
q3
c
16
H
H
RX −0.351
RX −0.351
RX −0.351
RX −0.351
ZZ (−0.829)
ZZ (−0.829)
0
RXX 0.244
1
0
RXX 0.244
1
ZZ (−0.0603)
ZZ (−0.0603)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
RX −0.351
RX −0.351
RX −0.351
RX −0.351
ZZ (−0.829)
ZZ (−0.829)
0
RXX 0.244
1
0
RXX 0.244
1
ZZ (−0.0603)
ZZ (−0.0603)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

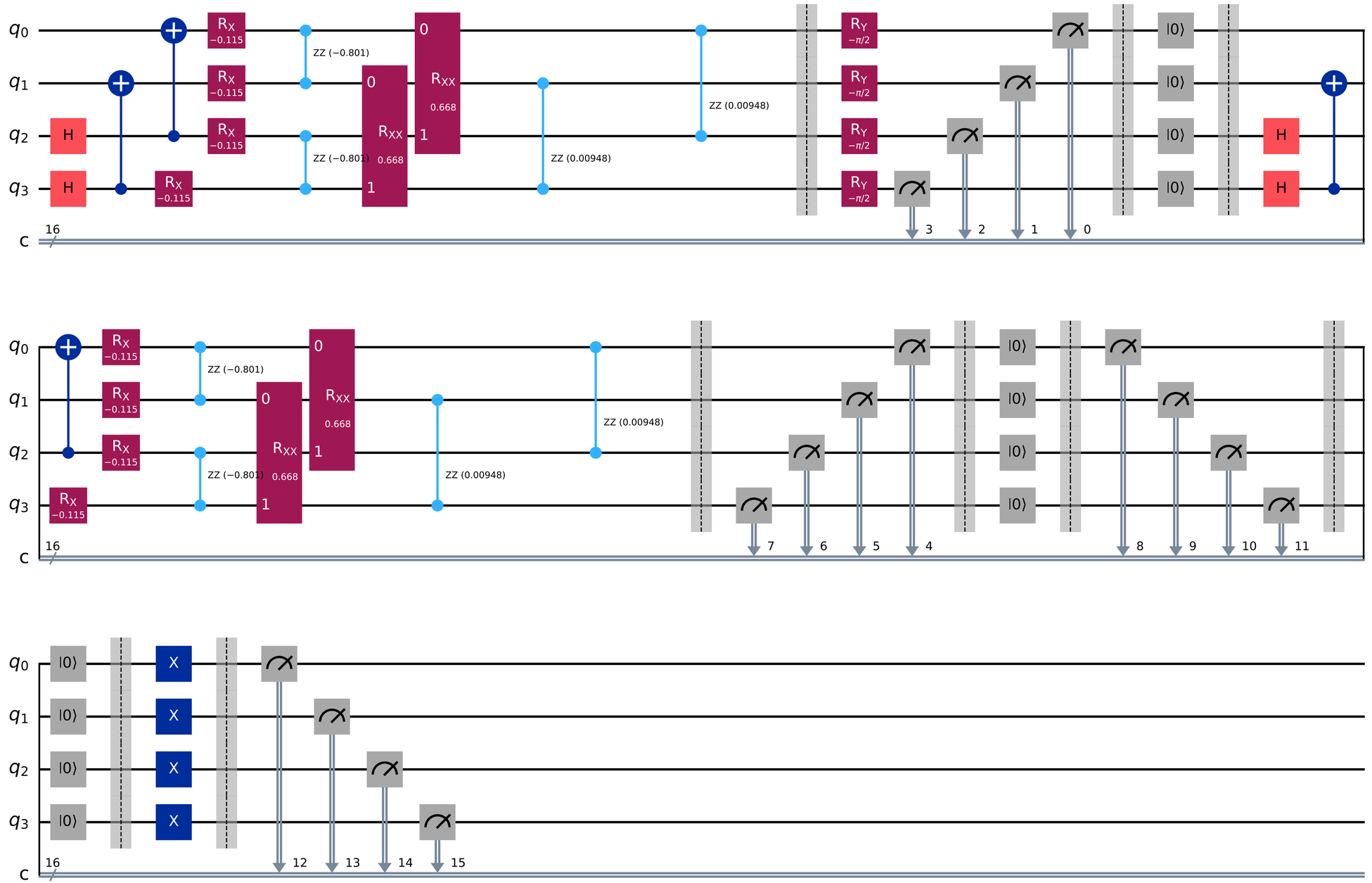

q0
q1
q2
q3
c
16
H
H
RX −0.153
RX −0.153
RX −0.153
RX −0.153
ZZ (−0.999)
ZZ (−0.999)
0
RXX 0.876
1
0
RXX 0.876
1
ZZ (−0.0477)
ZZ (−0.0477)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
RX −0.153
RX −0.153
RX −0.153
RX −0.153
ZZ (−0.999)
ZZ (−0.999)
0
RXX 0.876
1
0
RXX 0.876
1
ZZ (−0.0477)
ZZ (−0.0477)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

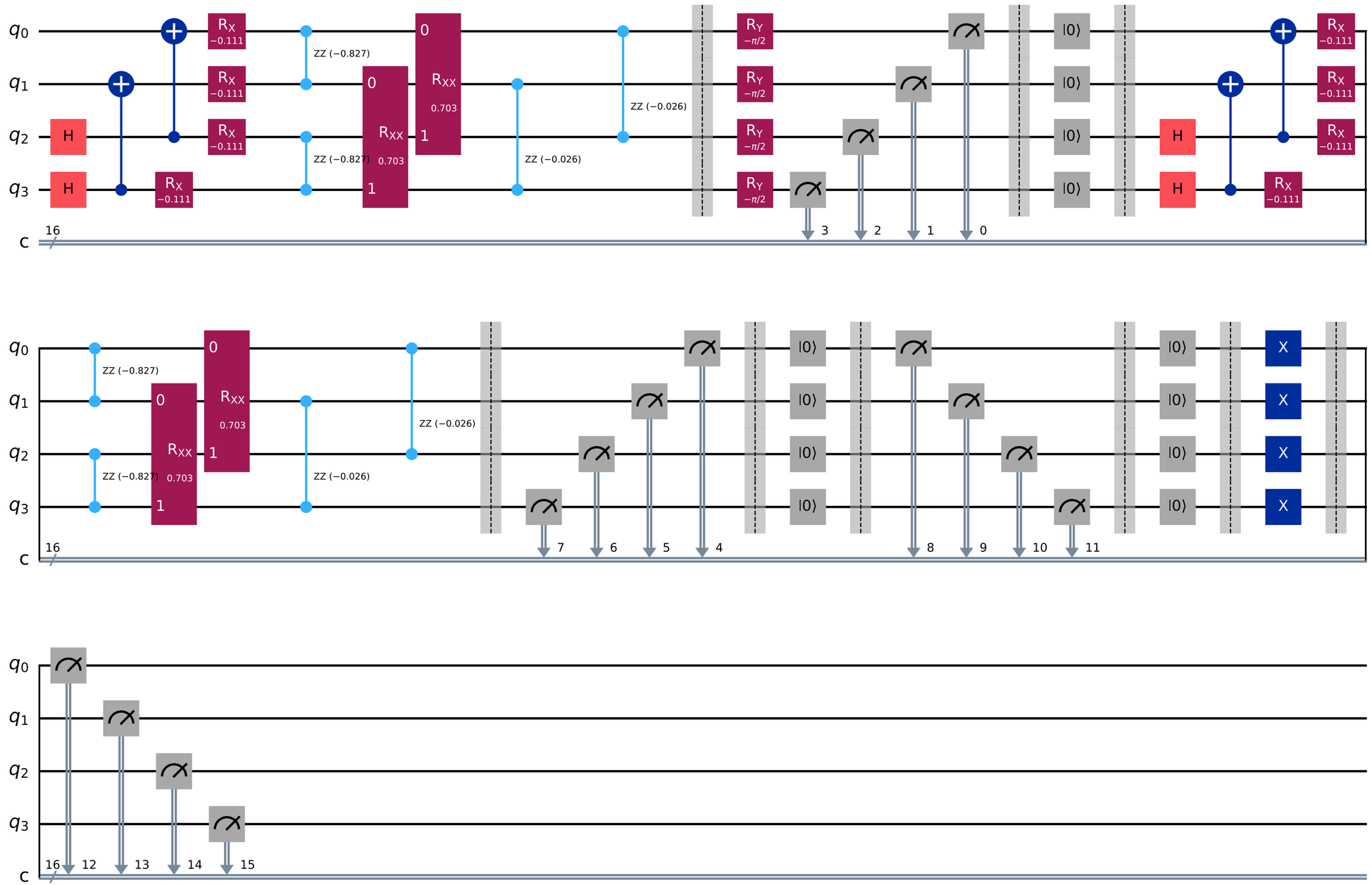

q0
q1
q2
q3
c
16
H
H
RX −0.0916
RX −0.0916
RX −0.0916
RX −0.0916
ZZ (−0.867)
ZZ (−0.867)
0
RXX
0.672
1
ZZ (−0.106)
ZZ (−0.106)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0916
RX −0.0916
RX −0.0916
RX −0.0916
ZZ (−0.867)
ZZ (−0.867)
ZZ (−0.106)
ZZ (−0.106)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

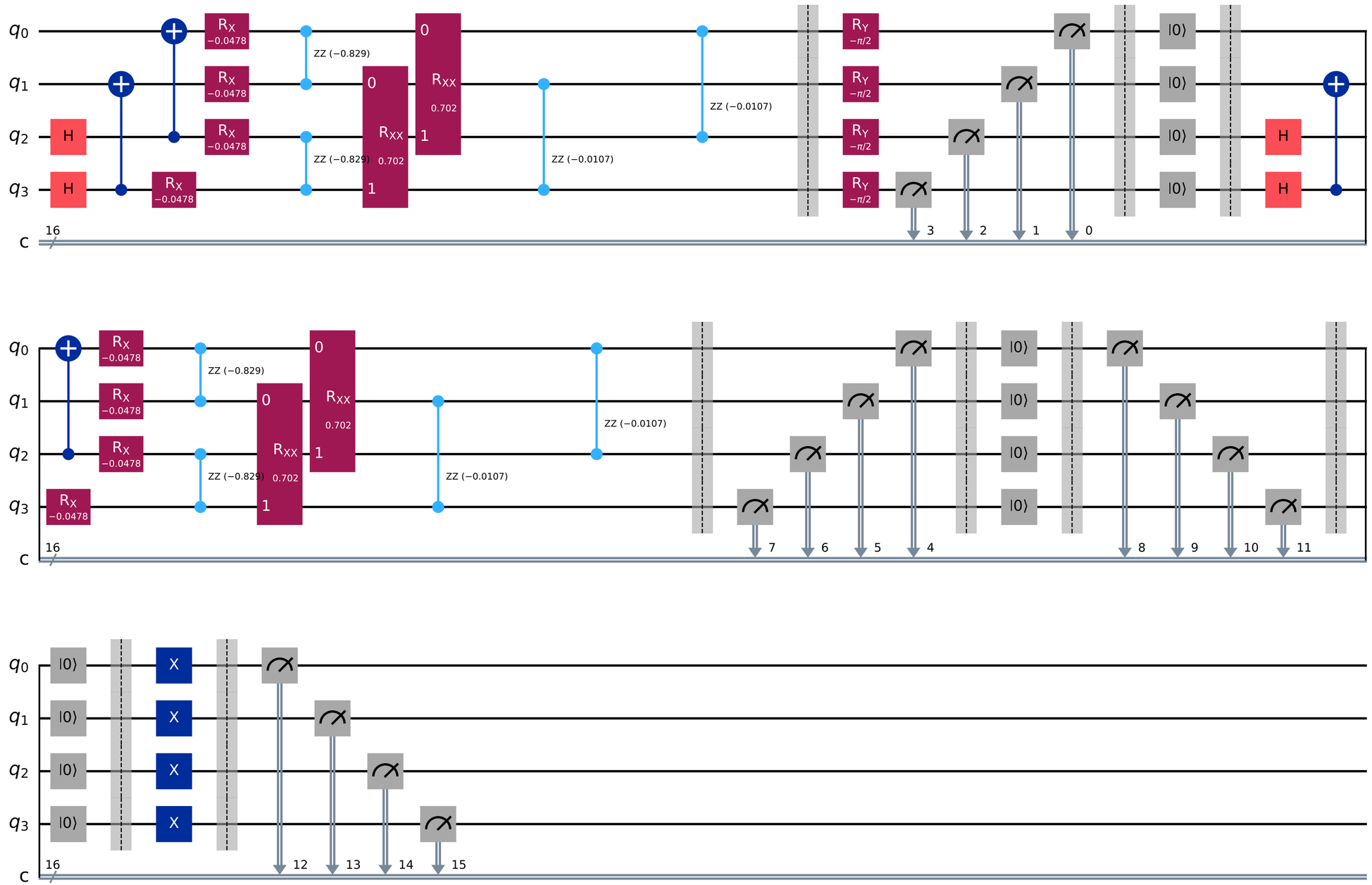

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0164
ZZ (−0.908)
ZZ (−0.0102)
R_{XX} 0.656
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

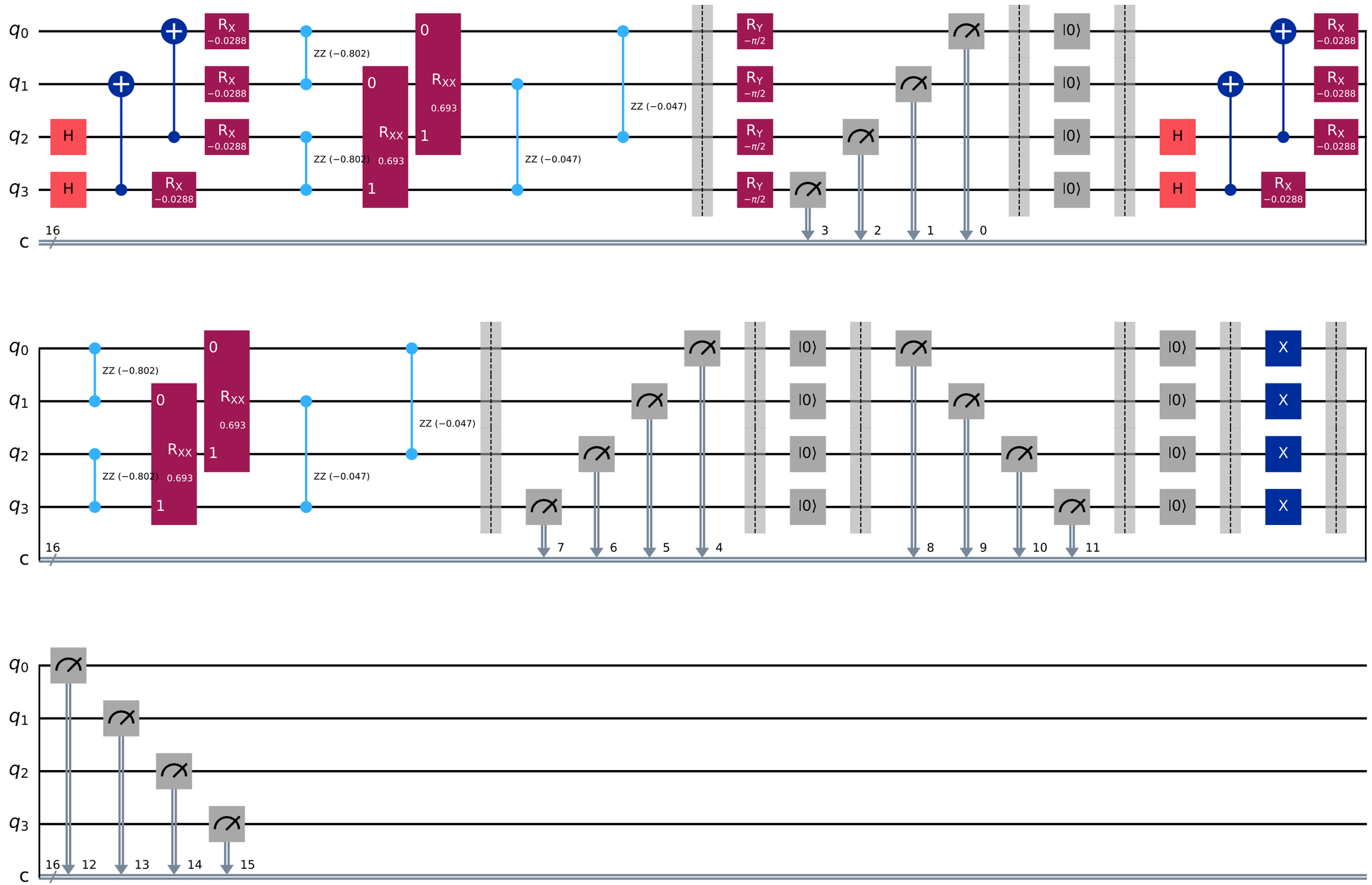

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.104
R_X −0.104
R_X −0.104
R_X −0.104
ZZ (−0.885)
ZZ (−0.885)
0
R_{XX} 0.657
1
0
R_{XX} 0.657
1
ZZ (−0.0446)
ZZ (−0.0446)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.104
R_X −0.104
R_X −0.104
R_X −0.104
ZZ (−0.885)
ZZ (−0.885)
0
R_{XX} 0.657
1
0
R_{XX} 0.657
1
ZZ (−0.0446)
ZZ (−0.0446)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

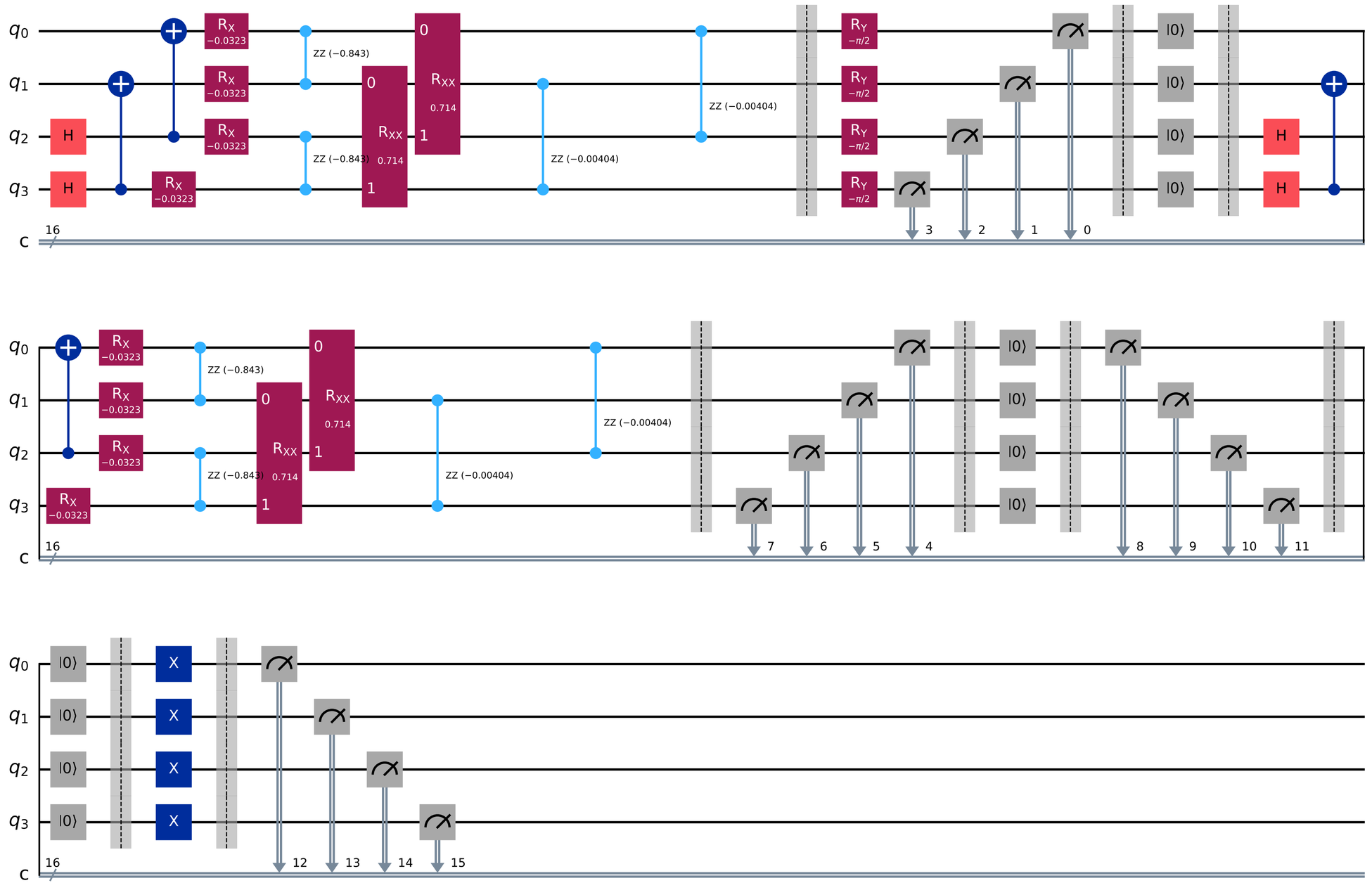

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0927
R_X −0.0927
R_X −0.0927
R_X −0.0927
ZZ (−0.823)
ZZ (−0.823)
0
R_{XX}
0.687
1
0
R_{XX}
0.687
1
ZZ (−0.0248)
ZZ (−0.0248)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0927
R_X −0.0927
R_X −0.0927
R_X −0.0927
ZZ (−0.823)
ZZ (−0.823)
0
R_{XX}
0.687
1
ZZ (−0.0248)
ZZ (−0.0248)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

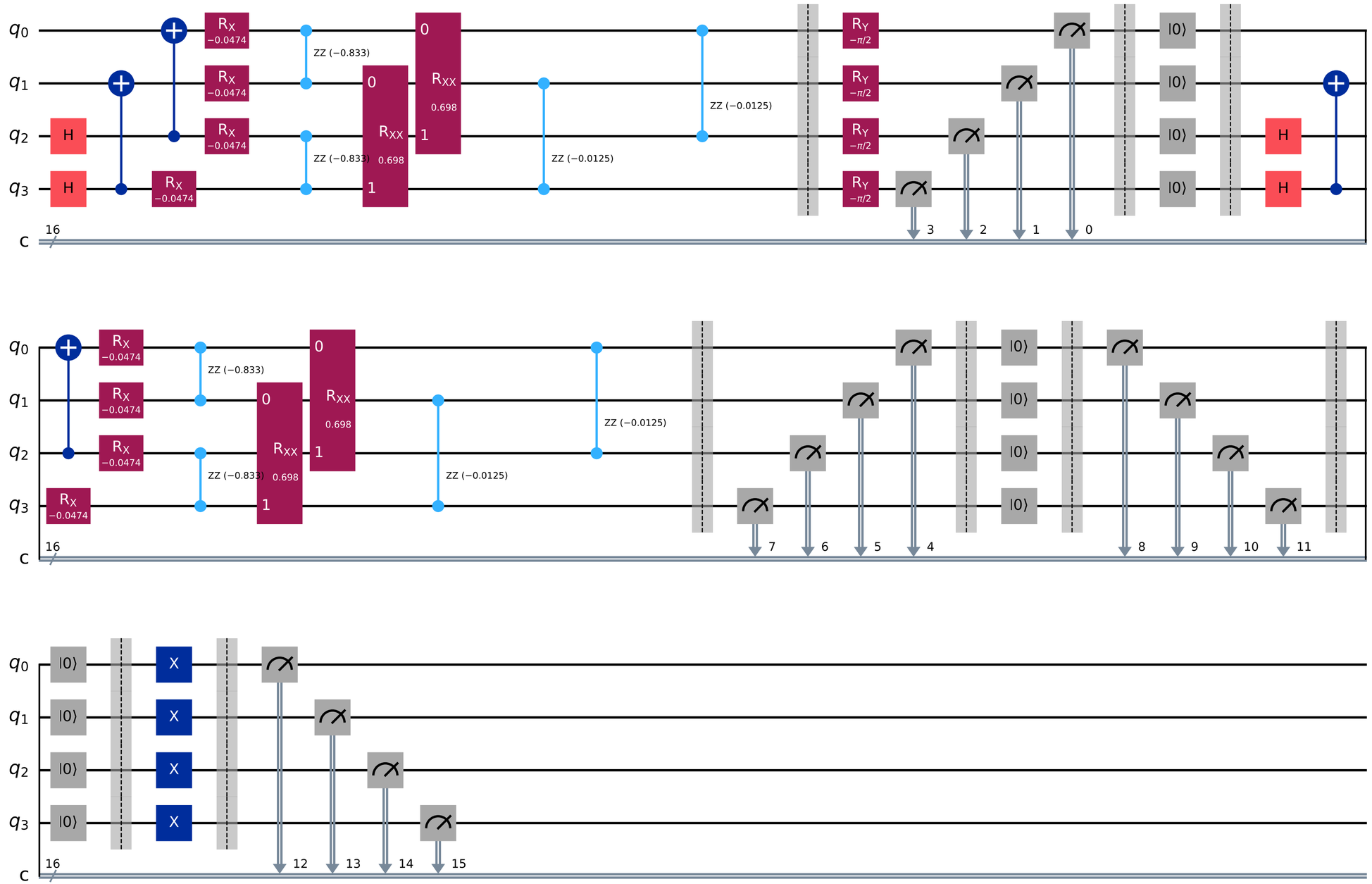

q0
q1
q2
q3
c
16
H
H
RX −0.115
RX −0.115
RX −0.115
RX −0.115
ZZ (−0.801)
ZZ (−0.801)
0
RXX 0.668
1
0
RXX 0.668
1
ZZ (0.00948)
ZZ (0.00948)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.115
RX −0.115
RX −0.115
RX −0.115
ZZ (−0.801)
ZZ (−0.801)
0
RXX 0.668
1
0
RXX 0.668
1
ZZ (0.00948)
ZZ (0.00948)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

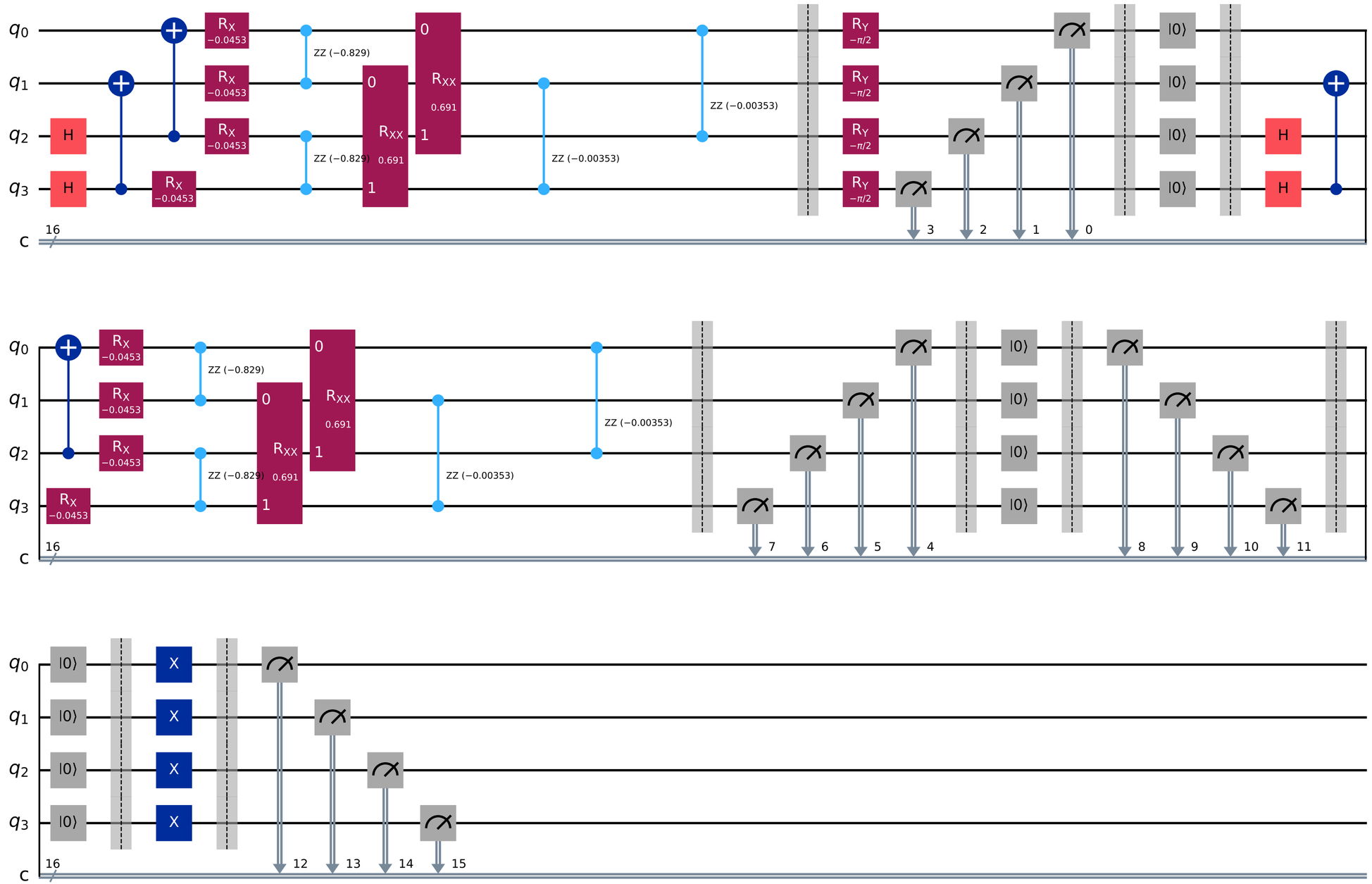

q0
q1
q2
q3
c
16
H
H
RX −0.111
RX −0.111
RX −0.111
RX −0.111
ZZ (−0.827)
ZZ (−0.827)
0
RXX 0.703
1
0
RXX 0.703
1
ZZ (−0.026)
ZZ (−0.026)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.111
RX −0.111
RX −0.111
RX −0.111
ZZ (−0.827)
ZZ (−0.827)
0
RXX 0.703
1
0
RXX 0.703
1
ZZ (−0.026)
ZZ (−0.026)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

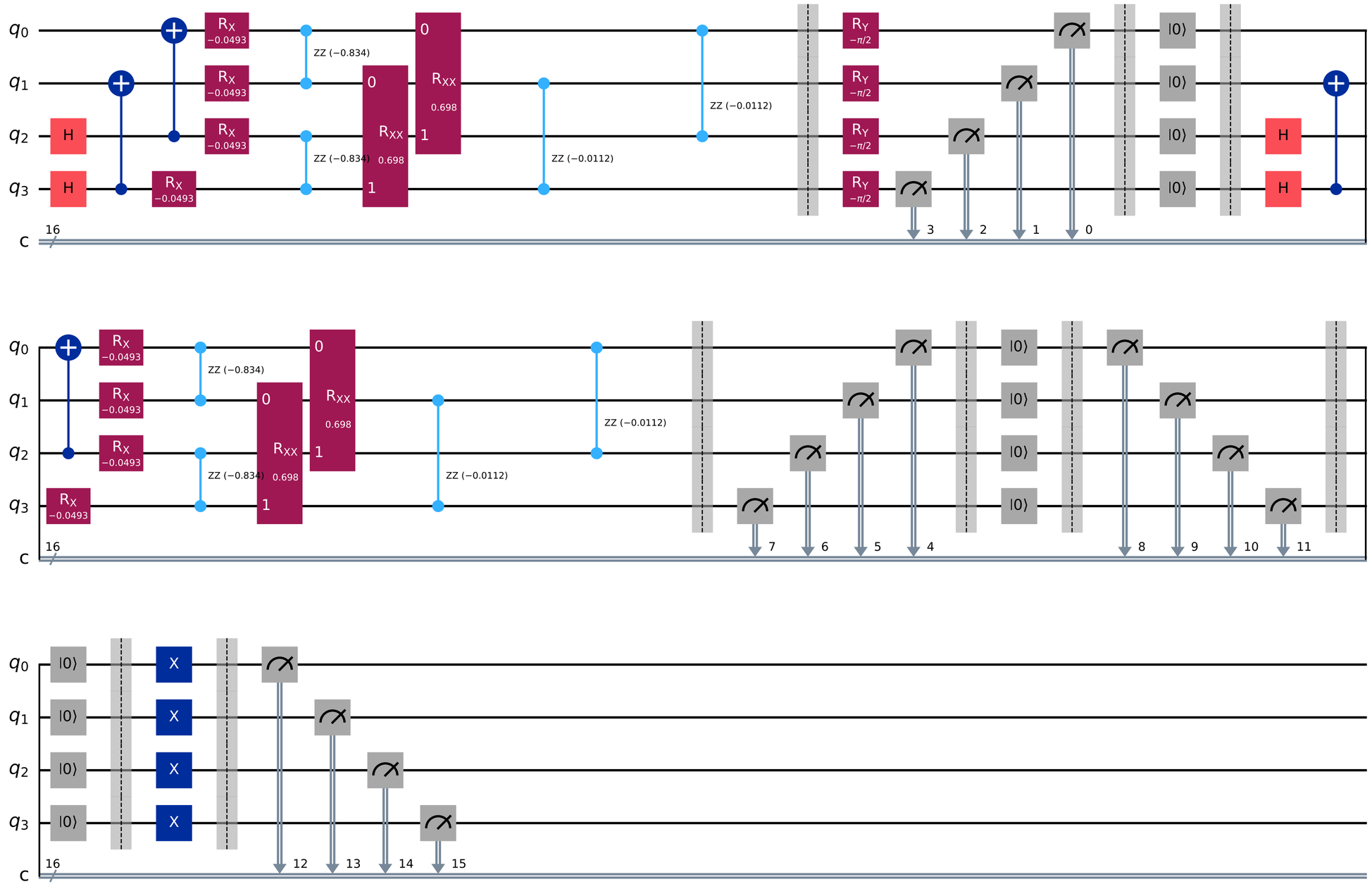

q0
q1
q2
q3
c
16
H
H
RX −0.0478
RX −0.0478
RX −0.0478
RX −0.0478
ZZ (−0.829)
ZZ (−0.829)
0
RXX 0.702
1
0
RXX 0.702
1
ZZ (−0.0107)
ZZ (−0.0107)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0478
RX −0.0478
RX −0.0478
RX −0.0478
ZZ (−0.829)
ZZ (−0.829)
0
RXX 0.702
1
ZZ (−0.0107)
ZZ (−0.0107)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
12
13
14
15
X
X
X
X

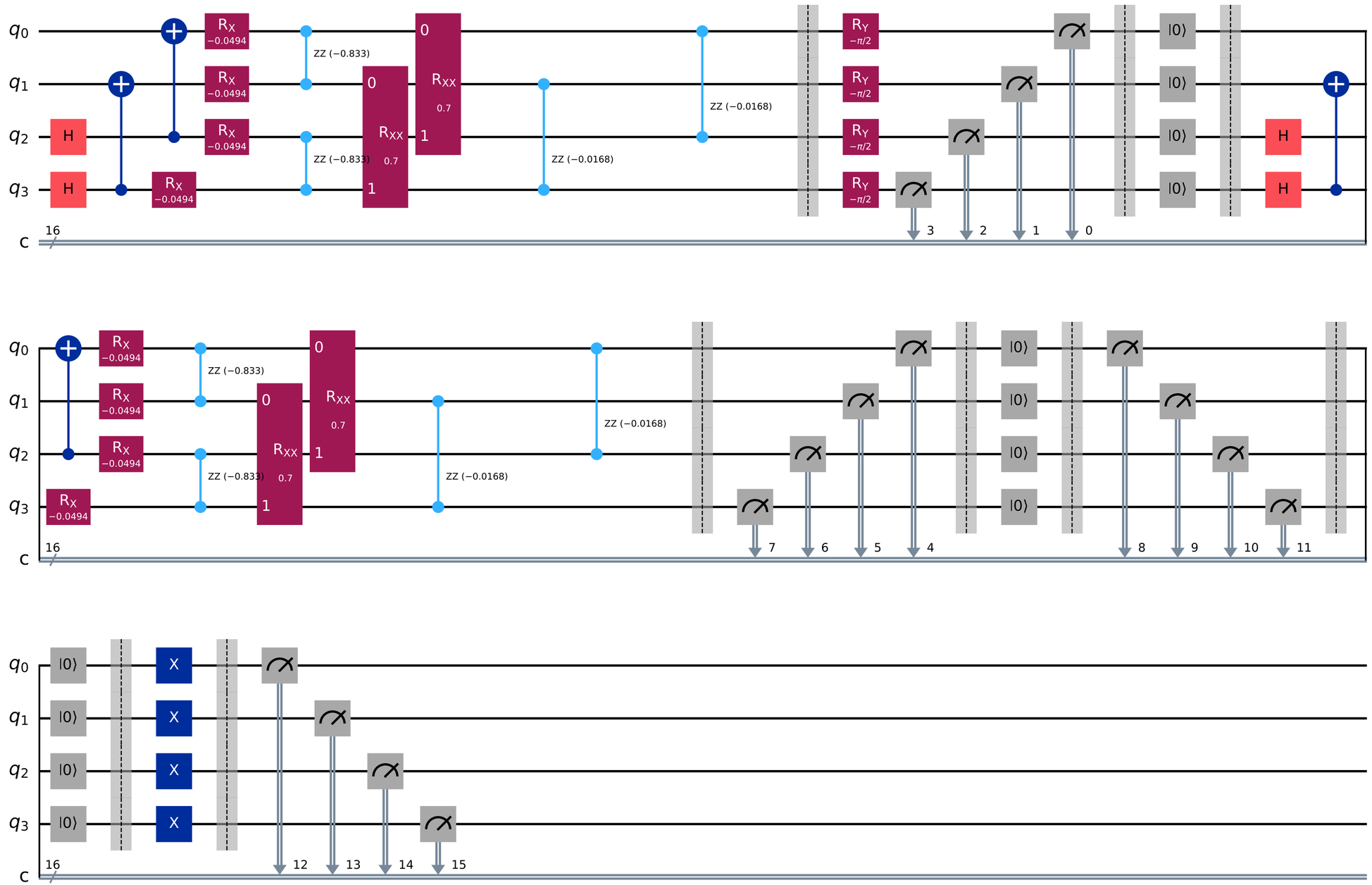

q0
q1
q2
q3
c
16
H
H
RX −0.0288
RX −0.0288
RX −0.0288
RX −0.0288
ZZ (−0.802)
ZZ (−0.802)
0
RXX 0.693
1
0
RXX 0.693
1
ZZ (−0.047)
ZZ (−0.047)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0288
RX −0.0288
RX −0.0288
RX −0.0288
ZZ (−0.802)
ZZ (−0.802)
0
RXX 0.693
1
0
RXX 0.693
1
ZZ (−0.047)
ZZ (−0.047)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

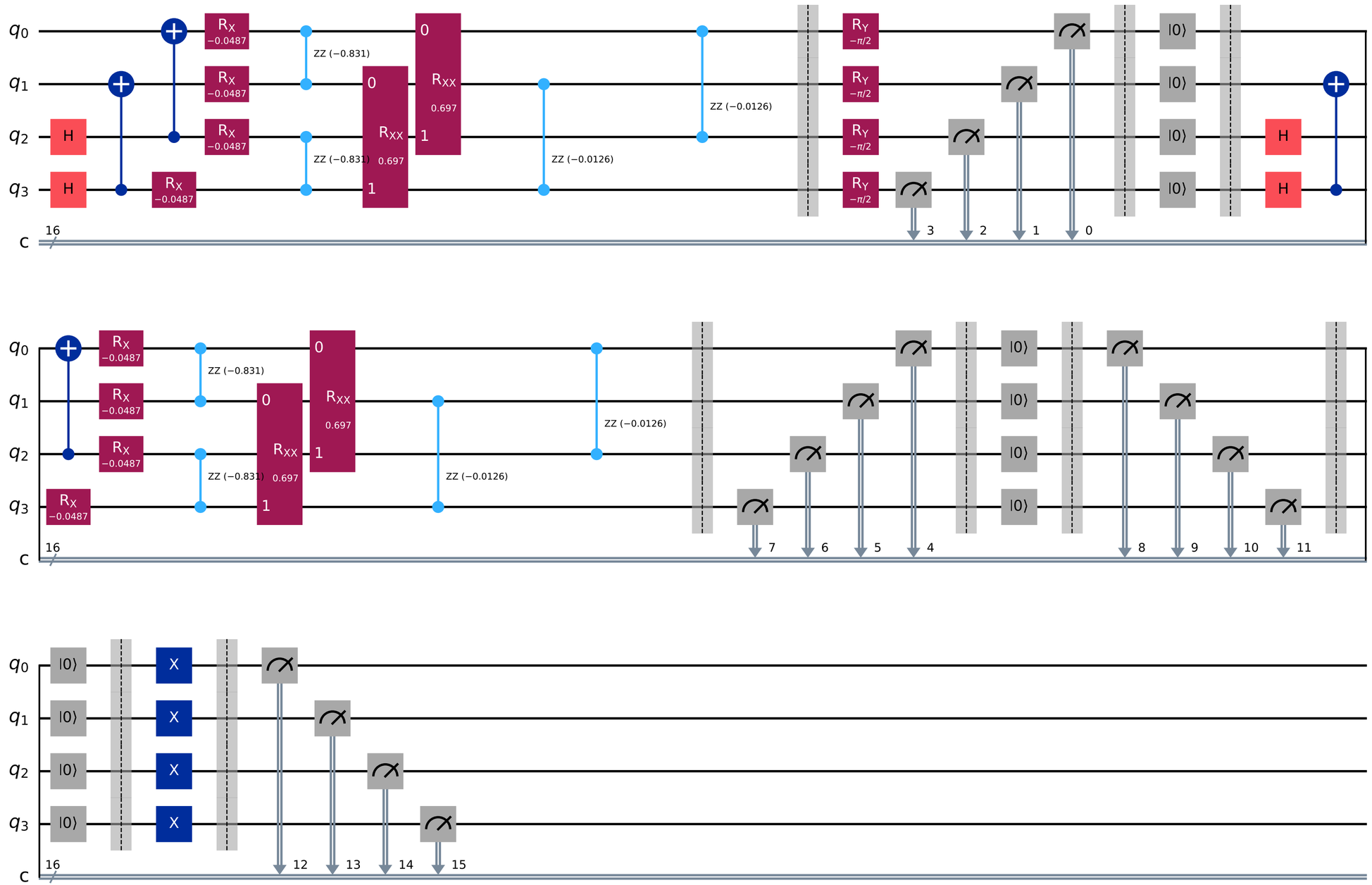

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0323
R_X −0.0323
R_X −0.0323
R_X −0.0323
ZZ (−0.843)
ZZ (−0.843)
0
R_{XX} 0.714
1
0
R_{XX} 0.714
1
ZZ (−0.00404)
ZZ (−0.00404)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.0323
R_X −0.0323
R_X −0.0323
R_X −0.0323
ZZ (−0.843)
ZZ (−0.843)
0
R_{XX} 0.714
1
0
R_{XX} 0.714
1
ZZ (−0.00404)
ZZ (−0.00404)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

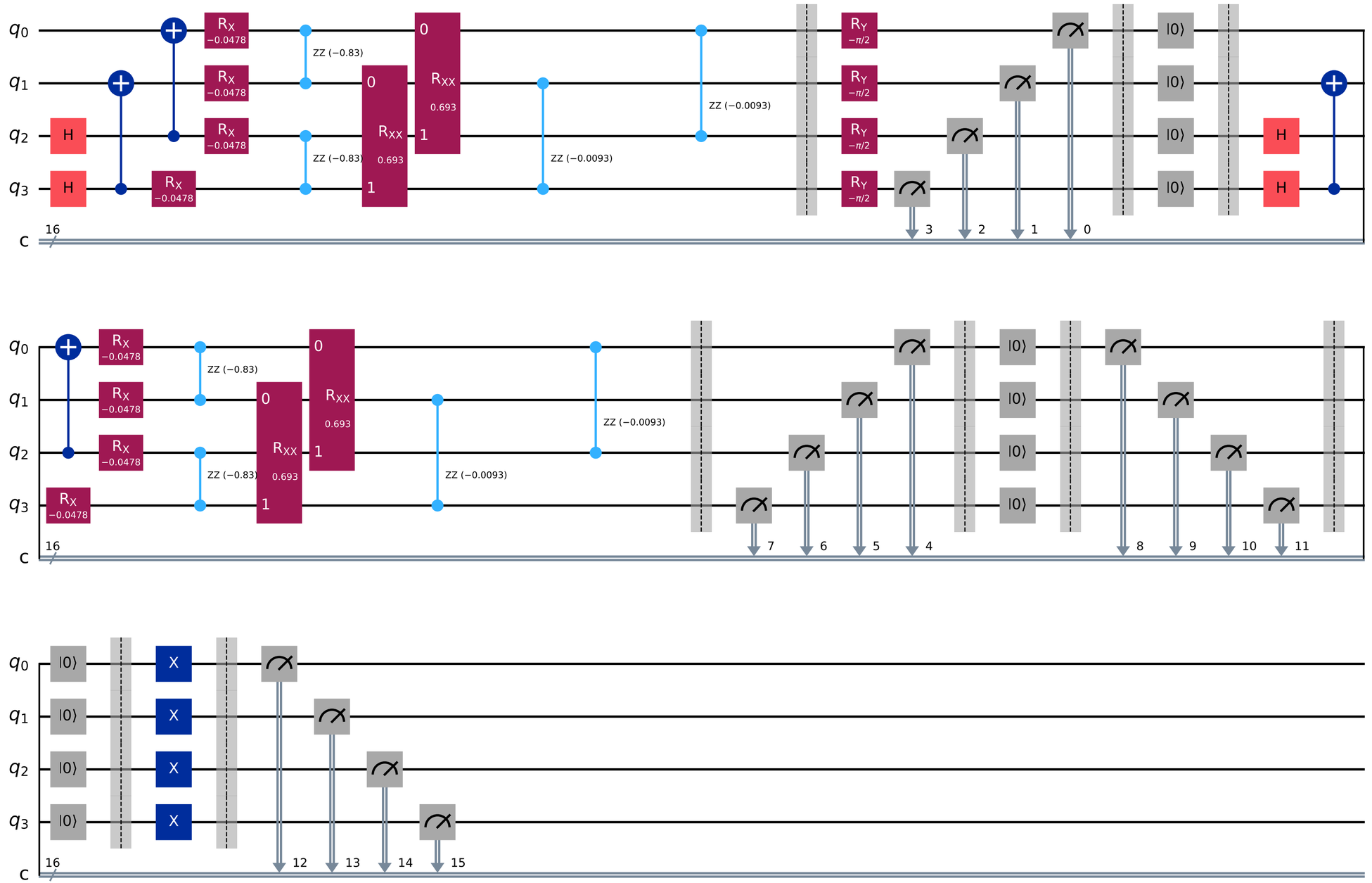

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0474
R_X −0.0474
R_X −0.0474
R_X −0.0474
ZZ (−0.833)
ZZ (−0.833)
0
1
R_{XX} 0.698
0
1
R_{XX} 0.698
ZZ (−0.0125)
ZZ (−0.0125)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.0474
R_X −0.0474
R_X −0.0474
R_X −0.0474
ZZ (−0.833)
ZZ (−0.833)
0
1
R_{XX} 0.698
0
1
R_{XX} 0.698
ZZ (−0.0125)
ZZ (−0.0125)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

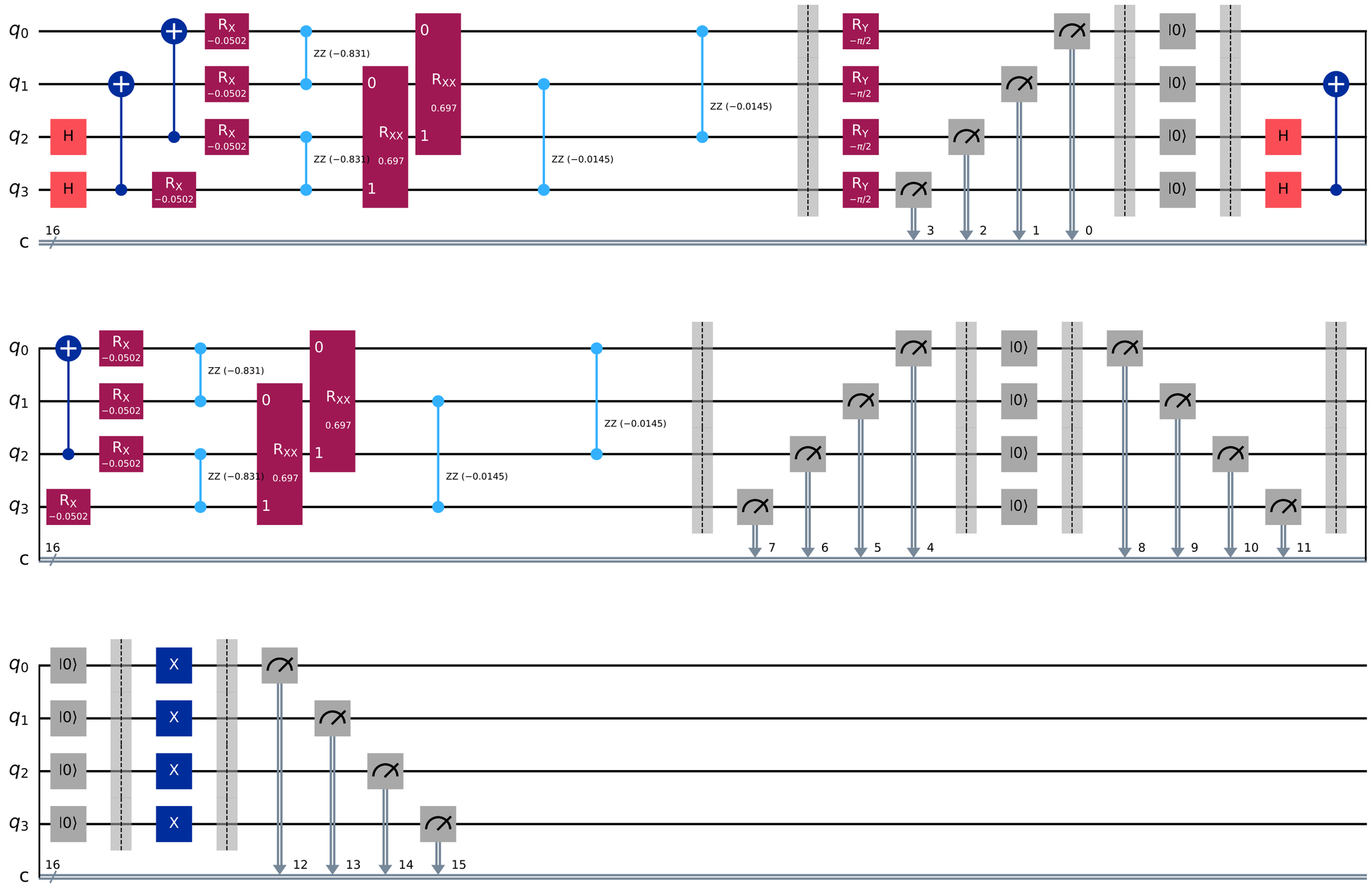

q0
q1
q2
q3
c
16
H
H
RX −0.0453
RX −0.0453
RX −0.0453
RX −0.0453
ZZ (−0.829)
ZZ (−0.829)
0
RXX 0.691
1
0
RXX 0.691
1
ZZ (−0.00353)
ZZ (−0.00353)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
RX −0.0453
RX −0.0453
RX −0.0453
RX −0.0453
ZZ (−0.829)
ZZ (−0.829)
0
RXX 0.691
1
0
RXX 0.691
1
ZZ (−0.00353)
ZZ (−0.00353)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

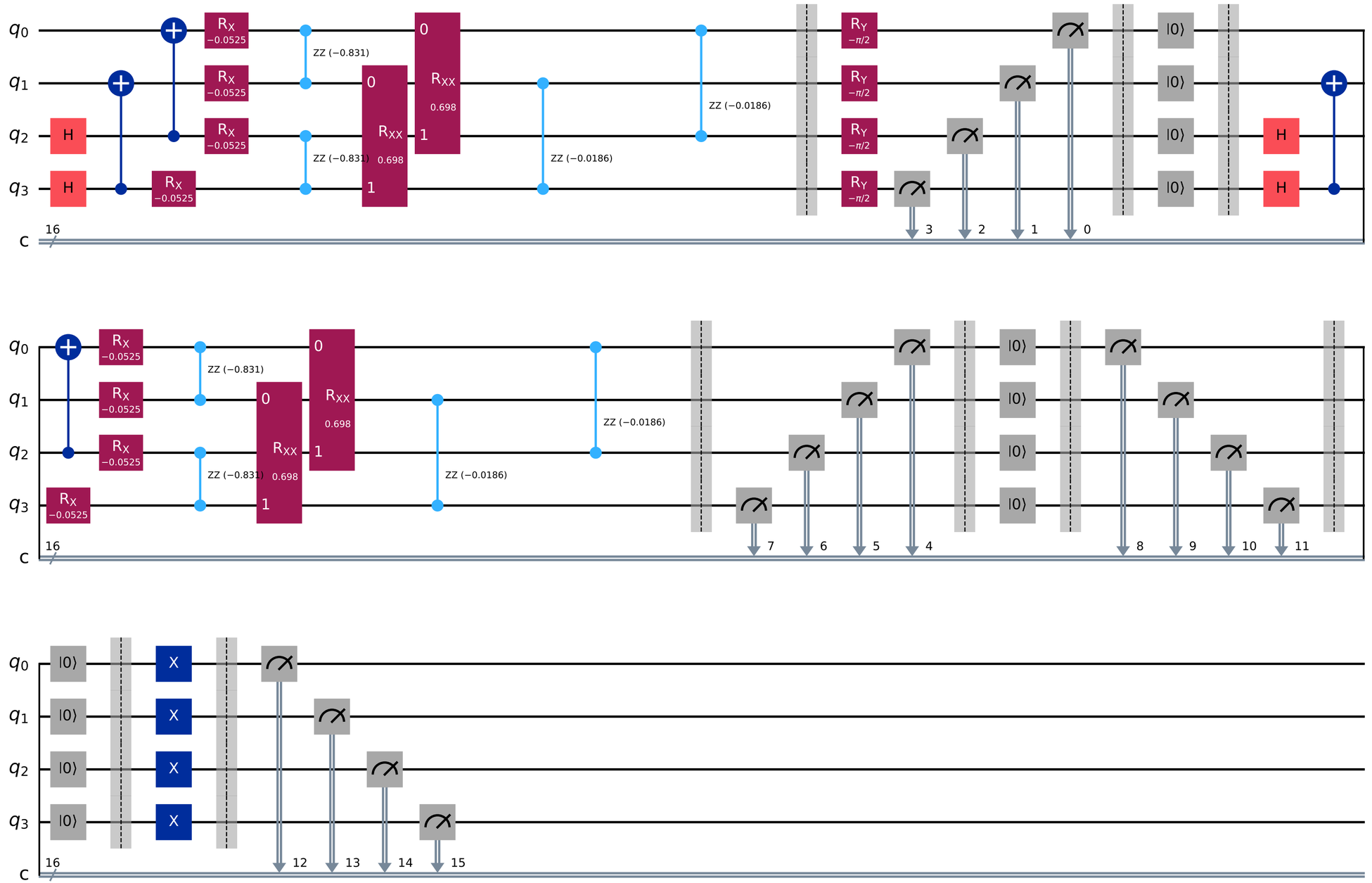

q0
q1
q2
q3
c
16
H
H
RX −0.0493
RX −0.0493
RX −0.0493
RX −0.0493
ZZ (−0.834)
ZZ (−0.834)
0
RXX 0.698
1
0
RXX 0.698
1
ZZ (−0.0112)
ZZ (−0.0112)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0493
RX −0.0493
RX −0.0493
RX −0.0493
ZZ (−0.834)
ZZ (−0.834)
0
RXX 0.698
1
0
RXX 0.698
1
ZZ (−0.0112)
ZZ (−0.0112)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

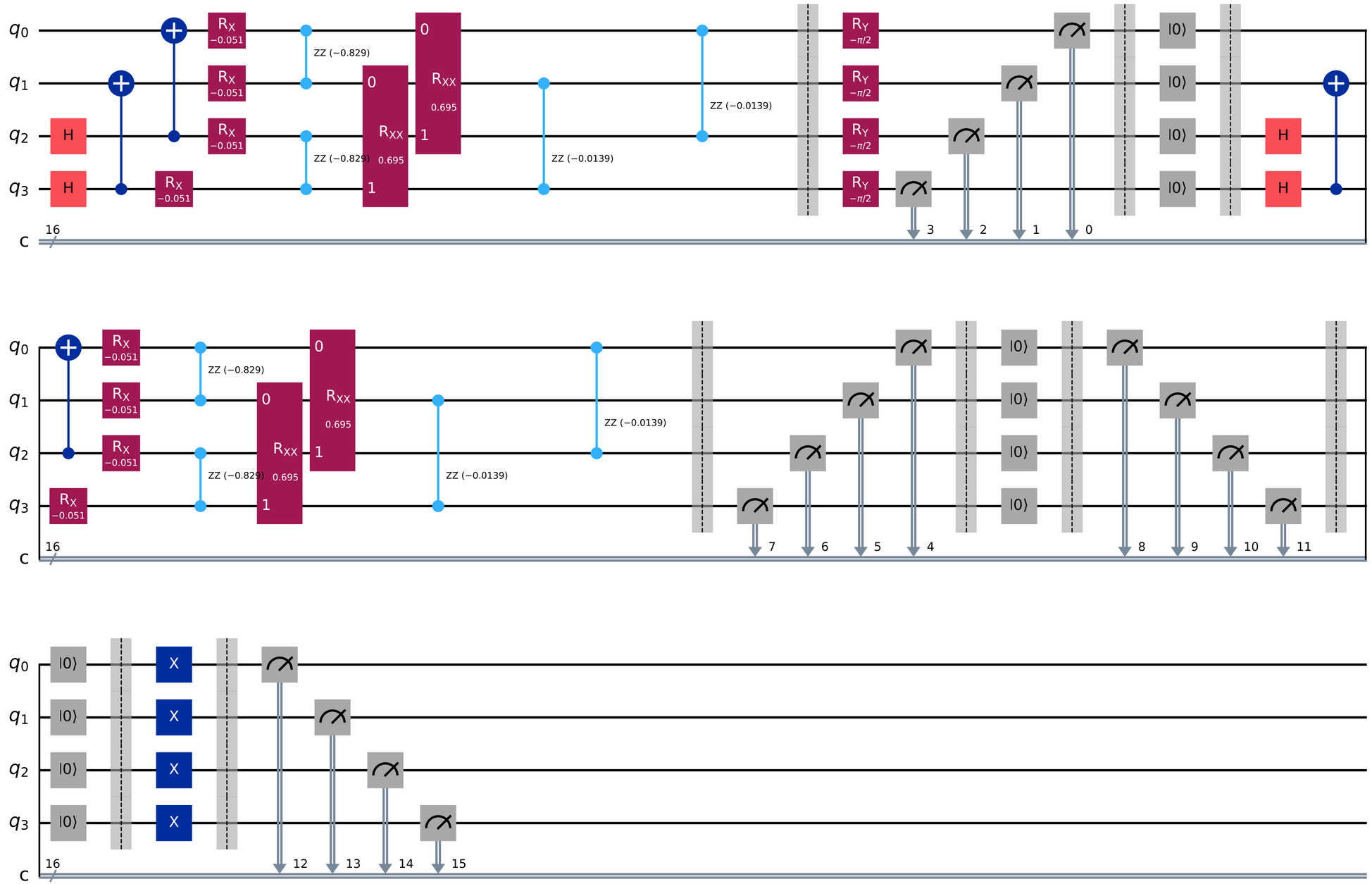

q0
q1
q2
q3
c
16
H
H
RX −0.0494
ZZ (−0.833)
RXX 0.7
ZZ (−0.0168)
RY −π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X

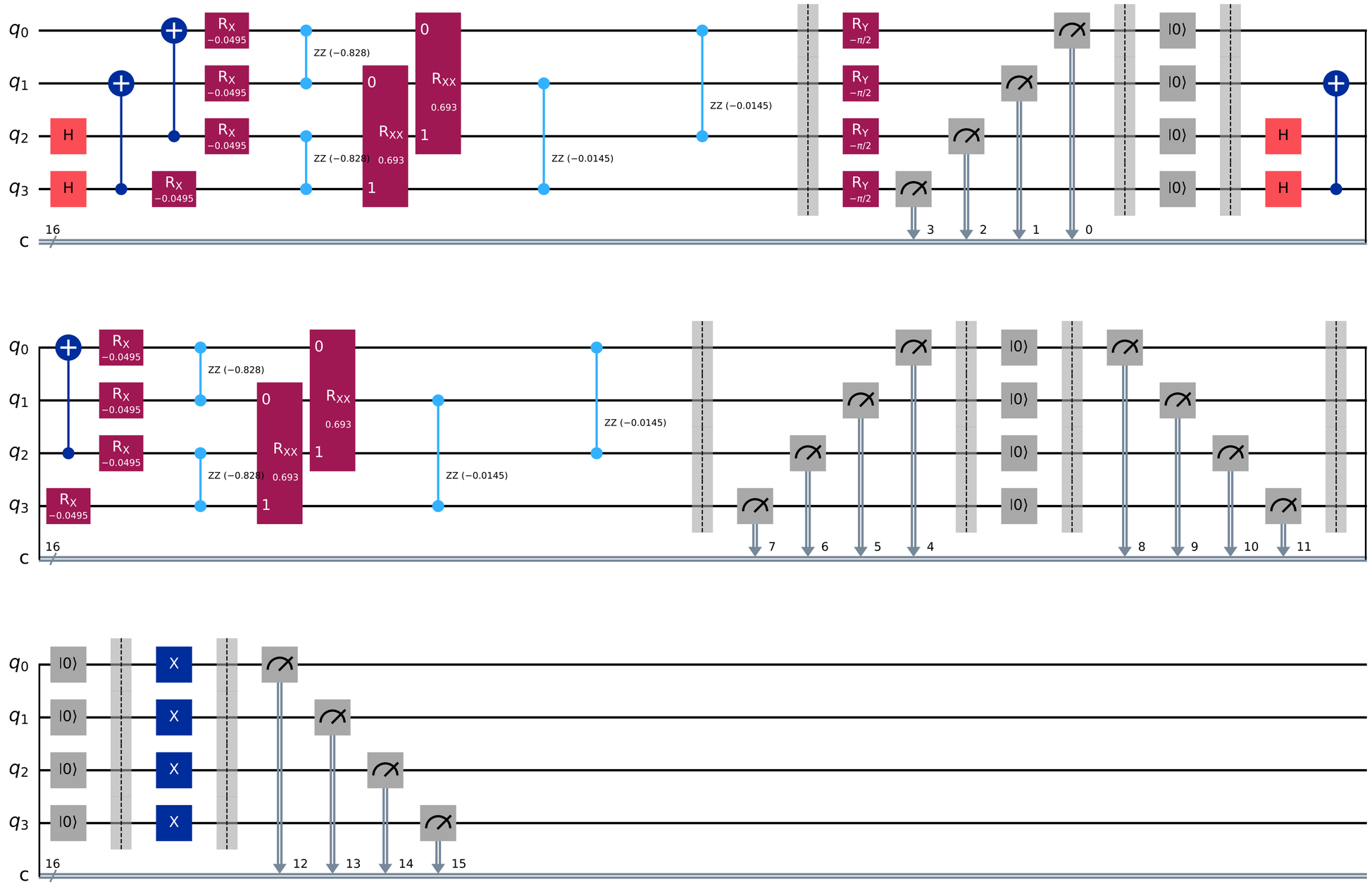

q0
q1
q2
q3
c
16
H
H
RX −0.0487
ZZ (−0.831)
0
RXX 0.697
1
ZZ (−0.0126)
RY −π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

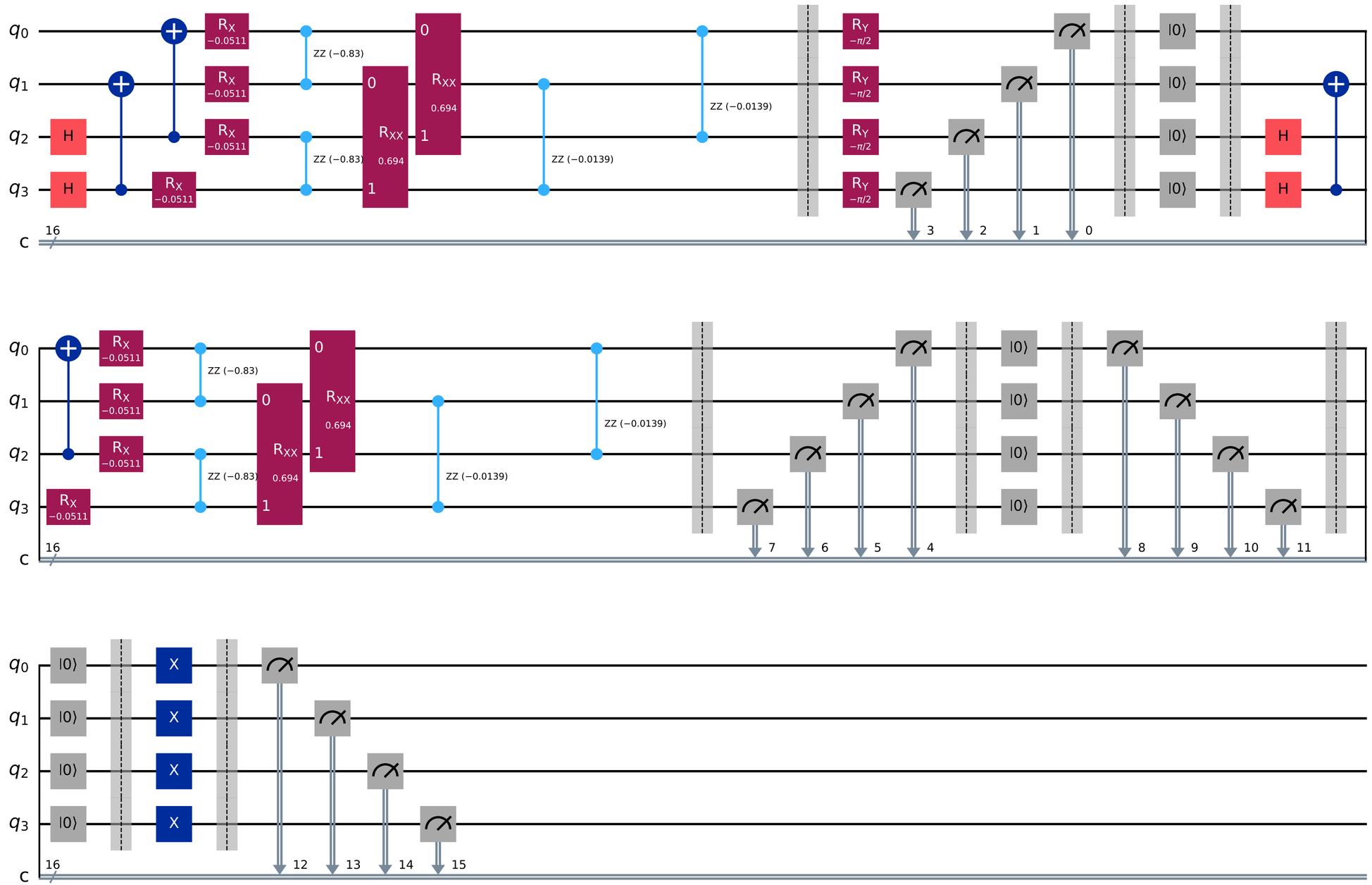

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0478
R_X −0.0478
R_X −0.0478
R_X −0.0478
ZZ (−0.83)
ZZ (−0.83)
0
1
R_{XX} 0.693
0
1
R_{XX} 0.693
ZZ (−0.0093)
ZZ (−0.0093)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.0478
R_X −0.0478
R_X −0.0478
R_X −0.0478
ZZ (−0.83)
ZZ (−0.83)
0
1
R_{XX} 0.693
0
1
R_{XX} 0.693
ZZ (−0.0093)
ZZ (−0.0093)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

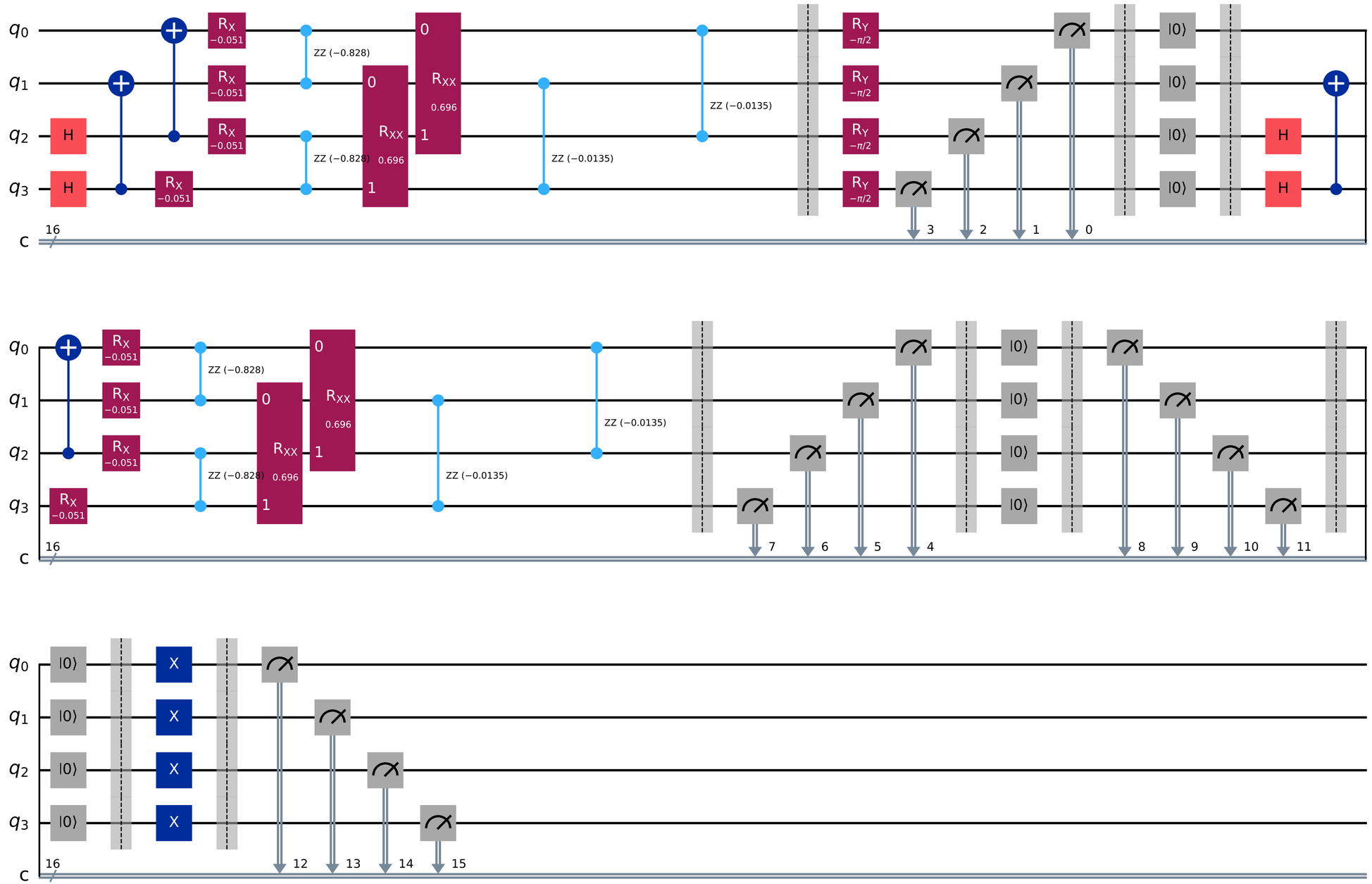

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0502
R_X −0.0502
R_X −0.0502
R_X −0.0502
ZZ (−0.831)
ZZ (−0.831)
0
R_{XX} 0.697
1
0
R_{XX} 0.697
1
ZZ (−0.0145)
ZZ (−0.0145)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.0502
R_X −0.0502
R_X −0.0502
R_X −0.0502
ZZ (−0.831)
ZZ (−0.831)
0
R_{XX} 0.697
1
0
R_{XX} 0.697
1
ZZ (−0.0145)
ZZ (−0.0145)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

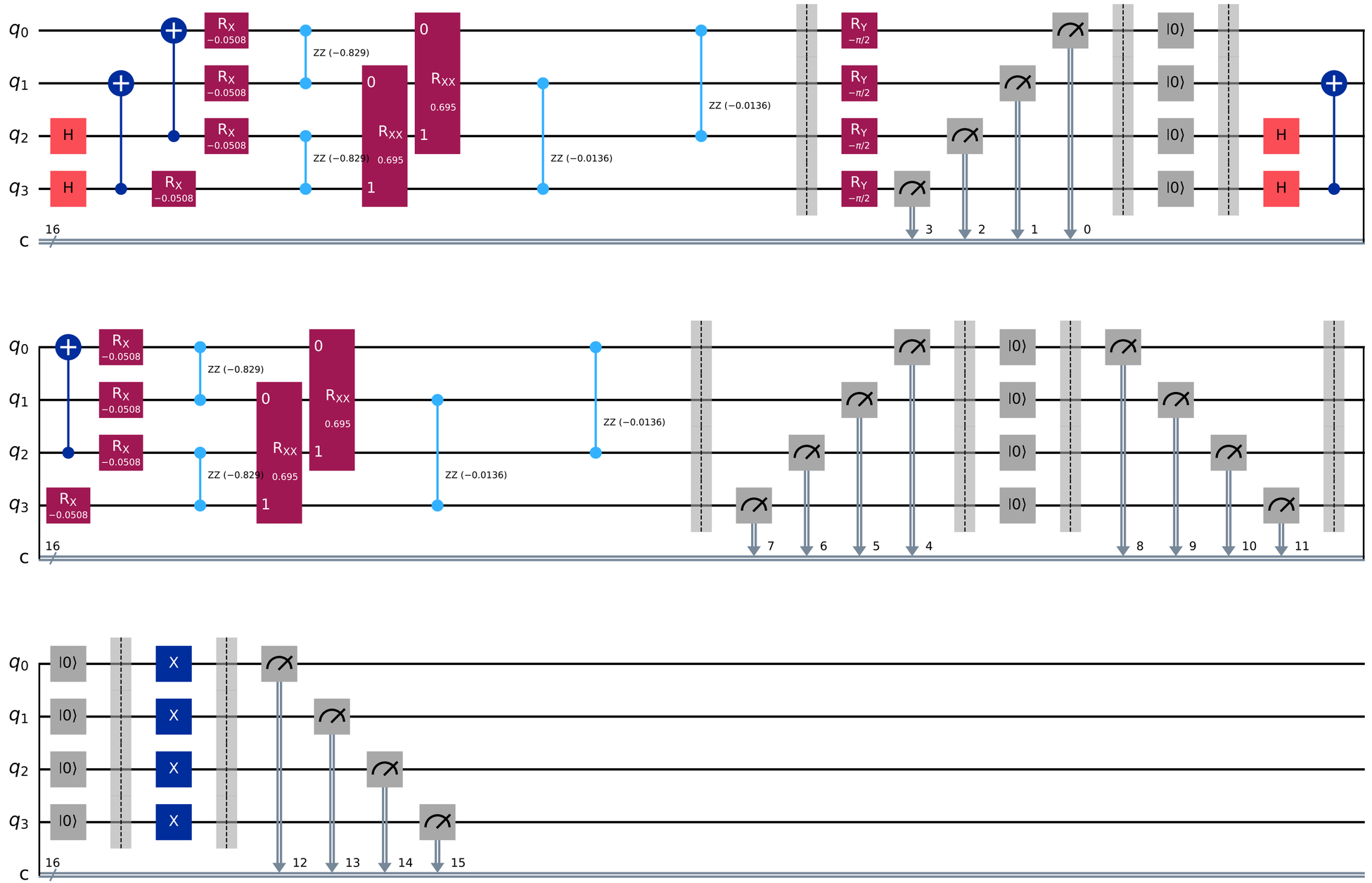

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0525
R_X −0.0525
R_X −0.0525
R_X −0.0525
ZZ (−0.831)
ZZ (−0.831)
0
R_{XX}
0.698
1
0
R_{XX}
0.698
1
ZZ (−0.0186)
ZZ (−0.0186)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.0525
R_X −0.0525
R_X −0.0525
R_X −0.0525
ZZ (−0.831)
ZZ (−0.831)
0
R_{XX}
0.698
1
0
R_{XX}
0.698
1
ZZ (−0.0186)
ZZ (−0.0186)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.051
ZZ (−0.829)
ZZ (−0.0139)
R_{XX} 0.695
R_Y −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0495
ZZ (−0.828)
ZZ (−0.828)
0
R_{XX}
0.693
1
ZZ (−0.0145)
ZZ (−0.0145)
R_Y −π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X


q0
q1
q2
q3
c
16
H
H
RX −0.0511
RX −0.0511
RX −0.0511
RX −0.0511
ZZ (−0.83)
ZZ (−0.83)
0
RXX 0.694
1
0
RXX 0.694
1
ZZ (−0.0139)
ZZ (−0.0139)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
RX −0.0511
RX −0.0511
RX −0.0511
RX −0.0511
ZZ (−0.83)
ZZ (−0.83)
0
RXX 0.694
1
0
RXX 0.694
1
ZZ (−0.0139)
ZZ (−0.0139)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.051
RX −0.051
RX −0.051
RX −0.051
ZZ (−0.828)
ZZ (−0.828)
0
RXX 0.696
1
0
RXX 0.696
1
ZZ (−0.0135)
ZZ (−0.0135)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.051
RX −0.051
RX −0.051
RX −0.051
ZZ (−0.828)
ZZ (−0.828)
0
RXX 0.696
1
ZZ (−0.0135)
ZZ (−0.0135)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0508
R_X −0.0508
R_X −0.0508
R_X −0.0508
ZZ (−0.829)
ZZ (−0.829)
0
R_{XX} 0.695
1
0
R_{XX} 0.695
1
ZZ (−0.0136)
ZZ (−0.0136)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
3
2
1
0
R_X −0.0508
R_X −0.0508
R_X −0.0508
R_X −0.0508
ZZ (−0.829)
ZZ (−0.829)
0
R_{XX} 0.695
1
0
R_{XX} 0.695
1
ZZ (−0.0136)
ZZ (−0.0136)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15